

"SNe Ia twins" in the Hubble flow, and the determination of H_0

P. RUIZ-LAPUENTE,^{1,2} A. QUINTANA-ESTELLÉS,^{1,2} J.I. GONZÁLEZ HERNÁNDEZ,^{3,4} AND A. PASTORELLO⁵

¹*Instituto de Física Fundamental, Consejo Superior de Investigaciones Científicas, c/. Serrano 121, E-28006, Madrid, Spain*

²*Institut de Ciències del Cosmos (UB-IEEC), c/. Martí i Franqués 1, E-08028, Barcelona, Spain*

³*Instituto de Astrofísica de Canarias, E-38200 La Laguna, Tenerife, Spain*

⁴*Universidad de La Laguna, Dept. Astrofísica, E38206 La Laguna, Tenerife, Spain*

⁵*INAF - Osservatorio Astronomico di Padova, Vicolo dell'Osservatorio 5, 35122 Padova, Italy*

ABSTRACT

We have applied our approach of using "SNe Ia twins" in the Hubble flow to obtain distances to SNe Ia at $z > 0.015$ and derive H_0 . Our results, taking a single step between the low z domain and the Hubble flow, validate the three rung classical method. We find, however, that the full compilation of distances, both in Pantheon+ and in the Carnegie-Chicago Hubble Program (CCHP), contain some inaccurate values in the colors due to an underestimate of reddening by dust, or due to the adoption of not well-defined light curve declines. This produces odd individual values for H_0 from single SNe Ia. On the average, those erroneous estimates do not affect the mean value of H_0 , which is characterized by the bulk of well-modeled SNe Ia. Our sample of carefully addressed SNe Ia in the Hubble flow contains a dozen supernovae, for which the distances are determined with high accuracy. Three of these SNe Ia are of the Broad Line subtype and can be compared with SN 1989B in M66, a host galaxy with a unique convergence of the Cepheid distance determination and the Tip of the Red Giant Branch stars (TRGB) determination by the CCHP group. There is as well a very good agreement on the distances to NGC 7250 and NGC 5643 between those derived with Cepheids by SH0ES and those derived with the use of J-Asymptotic Giant Branch stars (JAGB stars) by the CCHP, which makes them very good anchors. The sample of 12 SNe Ia gives a value of $H_0 = 72.56 \pm 1.54$ (stat) ± 1.33 (sys) $\text{km s}^{-1} \text{Mpc}^{-1}$, when anchored in Cepheids, and of $H_0 = 72.20 \pm 1.53$ (stat) ± 1.33 (sys) $\text{km s}^{-1} \text{Mpc}^{-1}$, when anchored in JAGBs by the CCHP. We take a mean of the two values of H_0 as derived by the Cepheids and by JAGB (from the CCHP) and obtain $H_0 = 72.38 \pm 1.54$ (stat) ± 1.33 (sys) $\text{km s}^{-1} \text{Mpc}^{-1}$. Our findings confirm that the Hubble tension is real.

Keywords: Cosmology, Hubble constant, Supernovae, general; supernovae, Type Ia; SN 2011fe, SN 2013aa, SN 2017cbv, SN 2013dy, SN 2012bo, SN 2008bq, SN 2008bz, LSQ12fxd, SN 2008bf, SN 2007A, ASASSN-15db, LSQ14gov, SN 2007ca, SN 2001cn, SN 2008go, SN 1999ek

1. INTRODUCTION

Type Ia supernovae (SNe Ia) at high z led to the discovery of the acceleration of the Universe (Riess et al. 1998; Perlmutter et al. 1999) and dark energy. Large compilations of SNe Ia have recently suggested a tentative evolution in time of such unknown major component of the energy density of the Universe (Rubin et al. 2025; DESyr5, DESI⁶).

Apart from this use in cosmology, a puzzle emerged some years ago and still requires an explanation: the expansion rate of the Universe today as obtained with SNe Ia calibrated with Cepheids gives a value of H_0 higher than that measured from the Cosmic Microwave Background (CMB). The value of H_0 derived from the CMB is $67.4 \pm 0.5 \text{ km s}^{-1} \text{Mpc}^{-1}$ (Planck Collaboration 2020) and the latest *SH0ES* value of $H_0 = 73.3 \pm 0.9 \text{ km s}^{-1} \text{Mpc}^{-1}$ (Murakami et al. 2023) have now a discrepancy at the 5.7σ level or even at the 6σ level (Riess et al. 2025). The *SH0ES* group has now addressed three additional methods: their own Tip of the Red Giant Branch stars (TRGB) measurements, to be compared with that of the Carnegie Chicago Supernova Project (CCHP),

⁶ SH0ES stands for "Supernova H0 for the Equation of State of Dark Energy". DESyr5 for "Dark Energy Survey year 5", DESI for "Dark Energy Spectroscopic Instrument".

their own J-Asymptotic Giant Branch stars (JAGB) and the use of Mira stars to compare with Cepheids, in conjunction with SNe Ia in the Hubble flow from the Pantheon+ SN catalog (Scolnic et al. 2022). They obtain values of $72.1\text{--}73.3 \pm 1.8 \text{ km s}^{-1} \text{ Mpc}^{-1}$ (depending on methodology) (Li et al. 2025).

Such difference in H_0 , when comparing with the H_0 provided by the CMB, if confirmed, would stand as a challenge to the Λ CDM model (Di Valentino et al. 2025a,b). A number of suggestions have been made, including the need of new physics in the early Universe to make the *Planck* value compatible with that derived by methods involving low- z astrophysical distance indicators such as Cepheids (see Di Valentino et al. 2021a,b, 2025a,b; Kamionkowski & Riess 2023; Schöneberg et al. 2021; Mörtzell & Dhawan 2018, Poulin et al. 2025, for reviews).

Nowadays, several other methods are giving values of H_0 around $74\text{--}72 \text{ km s}^{-1} \text{ Mpc}^{-1}$: the sample of distances obtained by the *Surface Brightness Fluctuations (SBF)* method, dominated by early-type galaxies as required by the method, calibrated with TRGB, gives rise to an $H_0 = 73.8 \pm 0.7$ (stat) ± 2.3 (syst) $\text{km s}^{-1} \text{ Mpc}^{-1}$ (Jensen et al. 2025). Garnavich et al. (2023) had used as well the SBF method and found $H_0 = 74.6 \pm 0.9$ (stat) ± 2.7 (syst) $\text{km s}^{-1} \text{ Mpc}^{-1}$. Galbany et al. (2023) used the peak of brightness in the infrared bands J and H from a sample of SNe Ia in the Hubble flow together with Cepheids by SH0ES as nearby calibrators and obtained H_0 of 72.3 ± 1.4 (stat) ± 1.4 (syst) $\text{km s}^{-1} \text{ Mpc}^{-1}$ when using the J-band and 72.3 ± 1.3 (stat) ± 1.4 (syst) $\text{km s}^{-1} \text{ Mpc}^{-1}$ when using the H-band. The Expanding Photosphere method, using SNe II, which does not rely in any nearby calibrator, gives 74.9 ± 1.9 (stat) $\text{km s}^{-1} \text{ Mpc}^{-1}$ (Vogl et al. 2024).

While new methods are now providing the above quoted values for H_0 , the debate on obtaining H_0 using Cepheids, TRGB or JAGB stars distances to calibrate Type Ia supernovae (SN Ia) persists. The SNe Ia calibration depends on three rungs: Rung 1 is the calibration of TRGB/Cepheids/JAGB stars with local anchors using geometrical methods, rung 2 refers to the calibration of the distance to SNe Ia host galaxies with TRGB/Cepheids/JAGB stars, and rung 3 is the overall calibration of SNe Ia in the Hubble flow, which is done in different ways but usually involves the derivation of a fiducial value for the absolute magnitude of SNe Ia and the final dependence of each particular SN Ia magnitude from such absolute magnitude and from factors such as rate of decline (hereafter called stretch) of the

light curve, color of the SN Ia at maximum and mass of its host galaxy. The two collaborations are making this step in different ways: Pantheon+ uses the SALT2 relation (Brout et al. 2022) while the CCHP uses a polynomial derivation (Freedman et al. 2019).

When using the TRGB and JAGB methods the two collaborations get different values of distances to the same galaxies and, therefore, different H_0 values from their use of these methods (see for instance Pantos & Perivolaropoulos, 2026; Ruiz-Lapuente & Gonzalez Hernández 2024: hereafter RLGH24).

Given the discrepancies between the two groups on H_0 , we decided to introduce a new approach (RLGH24). We address the Hubble tension issue with a method which uses the spectral evolution of SNe Ia that have not only similar stretch but identical spectral features along their lifetimes, i.e. "SNe Ia twins for life". The precision of the distance provided by those twins should be higher than for SNe Ia with just similar stretch. The total reddening could also be obtained.

The selection of twins is made of SNe Ia with a similar stretch, being then of similar luminosities, but in addition the "twinness factor" can make more precise the relative distance estimate, with a modulus error of 0.04 mag (RLGH24). As described in RLGH24, once the distance is determined by the MCMC, we position the twins at the same distance. We pass the spectrum through the B, V, R, and I filters and calculate the magnitude at B, V, R, and I. We do this for both SNe Ia twins. Then, we observe what the intrinsic dispersion of the twins might be, based on the magnitude difference. This yields a typical error of 0.04 mag at the filters.

In this work, we add another estimate of the accuracy of the method with an internal error, σ_{int} , from our MCMC calculations (see the Methodology section). It confirms an error ≤ 0.055 mag.

The anchors (/calibrators)⁷ used to obtain absolute distances should be galaxies where there is a confluence on the distance estimates obtained by various methods and groups.

Our method does not rely on many host galaxies of SNe Ia, but on a sample of 4 nearby host galaxies for which there is a very good agreement in the distance from the

⁷ We will call anchors or calibrators to the host galaxies in this first step of our method.

SH0ES and the CCHP collaborations. We use the distances to M101, NGC 7250, NGC 5643, M66 to anchor the twins. The uncertainty in the anchor is a part of the systematic error in the obtained H_0 . A difference of 0 mag indicates that the SH0ES and the CCHP collaborations agree in the anchor distances. However, an error of 0 mag is not reflecting the real uncertainty in the calibrators. We estimate it to be of 0.04 mag, as quoted typically in the most recent papers of the collaborations. Therefore, we have taken it to be 0.04 mag from anchoring of the calibrators for all SNe Ia.

The advantage to select this group of well measured galaxies is to avoid the whole range of discrepancies introduced by the present samples in rung 2 when applying different methods (Cepheids, TRGB, JAGB stars) by the two collaborations. The choice of these galaxies has been driven, as well, by the availability in number of SNe Ia twins in the Hubble flow of the kind found in these low- z galaxies.

The three-rung method (see Figure 6 in Riess et al. 2024) presents a sample of calibrators in rung 2 which looks far from being ideal: the calibrators point to individual H_0 values from 60 to 86 km s⁻¹ Mpc⁻¹. The calibrators in rung 2 are taken together with the sample of SNe Ia in the Hubble flow to define a fiducial absolute magnitude M_B for SNe Ia and the Hubble diagram. It seems clear that selecting particular subsamples of these galaxies for calibrating the SNe Ia would produce biases in H_0 . This fact has been identified as one of the causes of the divergent results for H_0 (Riess et al. 2024). In our method, the representation of the types of SNe Ia in the Hubble flow with a given intrinsic luminosity of SN Ia matches the luminosity of its twin SN Ia at low z . The uncertainty will come from the distance of the low z galaxies of each SN Ia type. We will avoid the selection effects arising from modeling a **diverse** SNe Ia family.

A crucial task now is to verify if the three-rung process, which culminates with distances given to the SNe Ia in the Hubble flow, is in accordance with the distances that we found by our more direct method. Our procedure is then a test of what is the most important delivery of the three rung method: the distances of SNe Ia in the Hubble flow and the value of H_0 .

The paper is organized as follows: In section 2, we describe the sample of nearby SNe Ia which will have twins in the Hubble flow. In section 3, we present the reasons to use those SNe Ia and their host galaxies as anchors. In section 4, we present our sample of SNe Ia in the Hubble flow. In section 5, we describe our method. In

section 6, we present the results on the distances of the SNe Ia. In this section, we discuss the distance moduli of SNe Ia and the H_0 obtained. We compare our results to the distances obtained by the Pantheon+ sample and by the CCHP. In section 7, we present a summary and give our conclusions.

2. THE USE OF TWIN SNE IA

In our previous paper (RLGH24), we tested the method in two SNe Ia from the same galaxy. Instead of μ , we were interested in $\Delta\mu$, the relative distance between the 2 twin SNe Ia, SN 2013aa and SN 2017cbv in NGC 5643. The results point to $\Delta E(B - V) = 0.0 \pm 0.0$ mag, as it should have been expected from the two SNe being in the same galaxy with similar conditions of negligible reddening in the host galaxy and same reddening by dust in our Galaxy, and a $\Delta\mu = 0.004 \pm 0.005$ mag in an early spectrum and $\Delta\mu = -0.023^{+0.008}_{-0.007}$ mag in a nebular one. Those correspond to a precision of 23 kpc from the early spectrum and around 100 kpc from the late one. The late phase has a larger error. The joint result from the two phases gives a distance difference compatible with 0 with $\Delta\mu = -0.005 \pm 0.004$ mag.

The preceding approach demonstrated the possibility to combine early and late phase information to obtain relative distances or differences in distance moduli between the hosts of SN Ia twins.

To estimate the magnitude accuracy that twins have in their spectral comparison, we calculated the difference in filters B, V, R, I filters of the twin spectra at the same phase once they are shifted to the same distance (this distance is calculated given the information provided by our emcee computation). The error estimate is of $\sigma = 0.04$ mag in all filters and all phases. This confirms the power of this method to disentangle errors in SNe Ia host galaxies distances provided by TRGB, Cepheids, JAGB.

Twin spectra of SNe Ia near maximum should have a similar shape of the pseudo-continuum and similar pseudo-equivalent widths (pWs) of the different lines. All that is required because otherwise it is not possible to have a difference of $\sigma = 0.04$ mag in magnitude in all filters.

Other approaches using the spectral diversity of SNe Ia twins in the context of improving the determination of the nature of dark energy were presented by Fakhouri et al. (2015) and Boone et al. (2016). In our case, we want to use the "twins" to determine the Hubble constant.

We follow the subclassification of SNe Ia according the *Branch classification* scheme (Branch et al. 2006), which has been set in more quantitative terms by the Carnegie Supernova Project. The *Branch classification* subdivides SNe Ia in four types: Core Normal (CN), Broad Line (BL), Shallow Silicon (SS) and Cool (CL). The main factor to be taken into account is the pseudo-equivalent widths (pWs) of the lines in the spectrum. Burrow et al.(2020) show how these classes are well separated in the pWs diagrams. The physical explanation of these subclasses is linked to the velocities and chemical composition of the SN Ia core which is revealed as the supernova photosphere recedes showing the inner ejecta. Those differences can be seen as four different subtypes with typical pseudo-equivalent widths (pWs) at various phases.

The spectra of the twins have very similar pseudo-equivalent widths (pWs) in the list of classifying lines of the subtypes (see Morrell et al. 2024). Those lines are pW1 (Ca H & K), pW2 (Si II λ 4130 Å), pW3 (Mg λ 4481; blended with Fe II), pW4 (Fe II at $\sim$ 4600 Å blended SII), pW5 (S II "W" $\sim$ 5400 Å), pW6 (Si II λ 5972 Å), pW7 (Si II λ 6355 Å), pW8 (Ca II IR Å). In particular, the line associated with Si II λ 5972 Å should be very similar in twin SNe Ia, with a discrepancy of less than 5%. This line correlates with stretch ($\Delta m(B)_{15}$, s_{BV}), and amongst SNe Ia of similar stretch, has to be almost the same for twins. In the pW6 over pW7 diagram, the twins fall in very close positions. If the equivalent widths are similar, the ratios of the two Si II at λ 5972 Å and λ 6355 Å, $\mathcal{R}_{SiII}$, should coincide. Thus, for the application of this method, both the pWs of the lines and the pseudo-continuum of the spectra have to be very similar between the twin SNe Ia. In particular, pW6 (Si II λ 5972 Å) and pW7 (Si II λ 6355 Å) should be similar within 5% (see RLGH24). The diagram of those pWs define the subtypes. In addition, pW6 has correlations with the stretch s_{BV} and $\Delta m(B)_{15}$.

We looked, in our method paper (RLGH24), at the abovementioned signs for the classification as twins of nearby SNe Ia. We need at least two phases to obtain a reliable result. (One phase is not enough).

In that previous paper (RLGH24), we used a single subtype of SNe Ia: the Core Normal (CN) subtype. In the present work we have extended it to the Broad Line (BL) subtype. We have found a good anchor of BL SNe Ia: SN 1989B in M66.

Previous research done within the core normal subtype started with a comparison between the twins SN

2013aa/SN 2017cbv, SN 2013aa and SN 2017cbv appeared in the outskirts of the same host galaxy, NGC 5643, so the reddening $E(B-V)$ by the host should be negligible. The reddening in our Galaxy in the direction of NGC 5643 is $(E - B)_{MW} = 0.15$ mag (Schlafly & Finkbeiner, 2011). SN 2013aa and SN 2017cbv have similar decline rates and B-peak magnitudes. The studies of these two SNe Ia also reveal similar characteristics in other aspects. Table 1 specifies the decline rates of those SNe Ia not only by $\Delta m_{15}(B)$ but also by s_{BV}^D (Burns et al. 2020). Fitting the B-V color curves with cubic splines, Burns et al (2020) find identical color-stretch $s_{BV}^D = 1.11$ for SN 2013aa and SN 2017cbv. This value is smaller (slower decline) than in typical SNe Ia. They also found similar values in the *Branch classification*, concerning the pseudo equivalent widths (pWs) of the lines of different elements. This proves that their interior was very alike in chemical composition. The use as anchor of the host galaxy of both SN 2013aa and SN 2017cbv NGC 5643 is justified given the converging values of distance moduli to this galaxy with the use of Cepheids by SH0ES and with JAGB stars by the CCHP.

In the previous paper, we found that SN 2013dy is practically equal at late phases to SN 2017cbv (there are no spectra of SN 2013aa at that phase to allow a comparison). A comparison shows that the light curve at late phases is slightly more luminous than in SN 2013aa, but coincides with SN 2017cbv. At very early phases it is similar to both SN 2017cbv and SN 2013aa though slightly redder. After maximum brightness, it is also identical to SN 2013aa. SN 2013dy has a the Galactic reddening $E(B-V)_{MW} = 0.14$ mag (Schlafly & Finkbeiner 2011), though there are indications that the total reddening might be 0.15 mag. There is the possibility, then, to use SN 2013dy as an anchor. The fact that this supernova was discovered a few hours after explosion and had a very intense photometric and spectroscopic follow up (Pan et al. 2015) enables to have many phases for comparison with their SNe Ia twins in the Hubble flow. There are, as well, converging values of distance moduli to the host galaxy of this SNIa, NGC 7250, with the use of Cepheids by SH0ES and with JAGB stars by the CCHP.

SN 2011fe, of the same *Branch subtype* as SN 2013aa, SN 2017cbv, and SN 2013dy, is also identified as a very good reference as both SH0ES and CCHP find similar distances to the host galaxy M101. SN 2011fe was discovered in its very early phase and was classified as a normal SN Ia (Nugent et al. 2011). At maximum light, the color was $(B_{max} - V_{max}) = -0.07 \pm 0.02$ mag.

Concerning the luminosity decline parameter of SN 2011fe, Burns et al (2023, private communication) measured a $\Delta m_{15}(B) = 1.07 \pm 0.006$ mag.

SN 2011fe is a faster and more underluminous SN Ia than SN 2017cbv and SN 2013aa. The difference between them is seen at both early and late phases. In fact, SN 2013aa and SN 2017cbv synthesized about 0.23 $M_{\odot}$ more ^{56}Ni than SN 2011fe. Studies by Jacobson-Galán et al. (2018) suggest that SN 2013aa and SN 2017cbv synthesized $0.732 \pm 0.151 M_{\odot}$ of ^{56}Ni . The mass of ^{56}Ni of SN 2011fe, as found by many authors, quoted in RLGH24, is $\sim 0.5 M_{\odot}$.

These four SNe Ia are the anchors for the CN SNe Ia in the Hubble flow. For BL SNe Ia in the Hubble flow, we can use as anchor SN 1989B. This supernova was studied in detail by Wells et al. (1994). We have been able to compare with our method significantly reddened SNe Ia in the Hubble flow with SN 1989B. One of the advantages of SN 1989B is that the distance determinations coincide using the Cepheids (Saha et al. 1999) and the TRGB by Hoyt et al. (2019) (see also Freedman et al 2019).

We present these SNe Ia in Table 1.

3. ON THE CONVERGENCE OF THE DISTANCE TO OUR ANCHORS

3.1. M101

Recently, a convergence has been achieved on the distance to M101 by the SH0ES group and the CCHP. This gives an extraordinary opportunity to be safe on the class of SN 2011fe-like supernovae. The distance to M101 is $\mu = 29.188 \pm 0.055$ mag for the *SH0ES* group using Cepheids (Riess et al. 2024) and $\mu = 29.18 \pm 0.04$ mag also from Cepheids by the CCHP in 2024 (Freedman et al. 2024) and $\mu = 29.151 \pm 0.04$ mag using TRGBs. In the most recent CCHP publication of the distance to this galaxy (Freedman et al. 2025), it is only slightly different.

There are other approaches giving slightly smaller distances to M101, like the Miras variables (Huang et al. 2024) and the Blue Supergiants (Bresolin et al 2025). All are within 1σ of the values from the SH0ES collaboration and the latest CCHP value quoted above. The JAGB stars distance to M101 from the CCHP (Freedman et al. 2025) is 29.208 ± 0.045 mag.

Table 1. SN 2013aa, SN 2017cbv, SN 2013dy SN 2011fe and SN 1989B

SN 2013aa		
RA, DEC ^a	14:32:33.881	-44:13:27.80
Discovery date ^a	...	2013-02-13
Phase (referred to maximum light) ^b	...	-7 days
Redshift ^c	...	0.004
E(B-V) _{MW} ^d	...	0.15 $\pm$ 0.06 mag
$m_B^{max\ b}$	...	11.094 $\pm$ 0.003 mag
$\Delta m(B)_{15}^b$	...	0.95 $\pm$ 0.01
Stretch factor s_{BV}^D ^b	...	1.11 $\pm$ 0.02
Phases of the spectra used	...	-2, 34, 42.07 days
SN 2017cbv		
RA, DEC ^e	14:32:34.420	-44:08:02.74
Discovery date ^e	...	2017-03-10
Phase (referred to maximum light) ^b	...	-18 days
Redshift ^c	...	0.004
E(B-V) _{MW} ^d	...	0.15 $\pm$ 0.06 mag
$m_B^{max\ b}$	...	11.11 $\pm$ 0.03 mag
$\Delta m(B)_{15}^b$	...	0.96 $\pm$ 0.02
Stretch factor s_{BV}^D ^b	...	1.11 $\pm$ 0.03
Phases of the spectra used	...	13.14 days
SN 2013dy		
RA, DEC ^f	22:18:17.599	+40:34:09.59
Discovery date ^f	...	2013-07-10
Phase (referred to maximum light) ^g	...	-18 days
Redshift ^h	...	0.0039
E(B-V) _{MW} ^d	...	0.15 mag
$m_B^{max\ i}$	...	13.229 $\pm$ 0.010 mag
$\Delta m(B)_{15}^j$	...	0.92 $\pm$ 0.006 mag
Stretch factor s_{BV}^D ^j	...	1.091 $\pm$ 0.03
Phases of the spectra used	...	3.7, 5 days
SN 2011fe		
RA, DEC ^a	14:03:05.810	+54:16:25.39
Discovery date ^k	...	2011-08-24
Phase (referred to maximum light) ^l	...	-20 days
Redshift ^m	...	0.0012
E(B-V) _{MW} ^d	...	0.008 mag
$m_B^{max\ n}$	...	9.983 $\pm$ 0.015 mag
$\Delta m(B)_{15}$	...	1.07 $\pm$ 0.06
Stretch factor s_{BV}	...	0.97 $\pm$ 0.002
Phases of the spectra used	...	4.65, 13 days
SN 1989B		
RA, DEC ^o	11:20:13.900	+13:00:19.00
Discovery date ^o	...	1989-01-30
Phase (referred to maximum light) ^p	...	-7 days
Redshift ^q	...	0.00242
E(B-V) _{MW} ^d	...	0.030 mag
$\Delta m(B)_{15}^q$	...	1.053 $\pm$ 0.11 mag
Stretch factor s_{BV}^q	...	0.954 $\pm$ 0.034 mag
Phases of the spectra used	...	-1, 3, 7, 9, 11 days
^a Waagen (2013).		
^c Parrent et al. (2013).		
^e Tartaglia et al. (2017).		
^g Zheng et al. (2013).		
ⁱ Pan et al. (2015).		
^k Richmond & Smith (2012)		
^m Burns (private communication, 2023).		
^p Evans (1989).		
^r Wells et al. (1994).		
^b Burns et al. (2020).		
^d Schlaflly & Finkbeiner (2011).		
^f Casper et al. (2013).		
^h Schneider et al. (1992).		
^j Uddin et al (2024)		
^l Cenko et al. (2011)		
ⁿ Yang et al. 2020.		
^q Morrell et al. (2024).		

3.2. NGC 5643

In the most recent papers by the SH0ES collaboration (years 2024–2025), the distances to NGC 5643 are given not only using the Cepheids method, but also the other methods used by the CCHP.

The values from SH0ES in the Table 2 from Li et al. (2024) include TRGB distances obtained with the JWST and they are compared with their Cepheids distance determination to the same galaxies. Their TRGB SH0ES values are not far from the Cepheids ones (only differing by ± 0.02 mag at most). The Cepheid distance modulus of reference for NGC 5643 is $\mu = 30.55 \pm 0.063$ mag, which is the same as in Riess et al. (2022). Their TRGB values slightly change if one uses different values for their smoothing $s = 0.10$ or $s = 0.05$. Fortunately, for NGC 5643 the difference is small ($\Delta\mu = 0.01$): the distance is 30.57 ± 0.06 mag, when using a smoothing of $s=0.10$, or 30.58 ± 0.06 mag when using a smoothing of $s=0.05$. The latest value from the TRGB method for NGC 5643 in 2025 obtained by the CCHP collaboration is $\mu = 30.643 \pm 0.066$ mag (Hoyt et al. 2025) but, in the compilation of Freedman et al. (2025), the error is of 0.071 mag. This value has changed often during the last year. The distance has gone in the last months from $\mu = 30.61 \pm 0.07$ mag (Freedman et al. 2024) to $\mu = 30.643 \pm 0.066$ mag (Hoyt et al. 2025). In 2021, the TRGB distance to NGC 5643 was significantly lower with a distance modulus of $\mu = 30.48 \pm 0.03$ (stat) ± 0.07 (sys) (Hoyt et al. 2021). There is a considerable difference in the TRGB values provided by SH0ES and by the CCHP, and even between the values provided by the CCHP in the last years.

Fortunately, there is an agreement on the distance to NGC 5643 between Cepheids used by SH0ES (Riess et al. 2022) and JAGB stars used by the CCHP (Freedman et al. 2025). As mentioned above the Cepheid distance value by SH0ES is $\mu = 30.55 \pm 0.06$ mag, and their TRGB value $\mu = 30.57/30.58 \pm 0.06$ mag. Whereas the TRGB value by the CCHP has been changing often, their JAGB value seems stable at $\mu = 30.582 \pm 0.038$ mag. (The JAGB value by SH0ES is 30.54 ± 0.04 , well in line with the previous values). We compare then our results using as anchor for this galaxy the Cepheids by SH0ES with the value anchored in the JAGB stars by the CCHP in section 6.

3.3. NGC 7250

In NGC 7250 there is good agreement between the distance obtained by SH0ES and the distance obtained by the CCHP with the JAGB stars.

The latest SH0ES measurement gives a distance of $\mu = 31.49 \pm 0.05$ mag using Cepheids measured with the JWST (Appendix 1 in Riess et al. 2025). The distance to NGC 7250 with Cepheids before the JWST had a 0.125 mag error. There is a very good agreement between the SH0ES collaboration and the CCHP collaboration using JAGB stars. The value by the SH0ES project is 31.59 ± 0.04 mag by Li et al. (2025) and 31.59 ± 0.02 (stat) ± 0.04 (sys) mag by the the CCHP project (Lee et al. 2025). This makes of NGC 7250 a good anchor, as central values coincide. On the contrary, the value using TRGB by the CCHP is $\mu = 31.629 \pm 0.034$ and it is different from the TRGB by SH0ES.

3.4. M66

This is a crucial SN Ia host whose distance has been measured with TRGB by Freedman et al. (2019) giving a distance modulus of $\mu = 30.22 \pm 0.04$ mag. In fact, this measurement seems to be based on Hoyt et al. (2019) TRGB measurement of the distance to M66, though the value quoted in this work is $\mu = 30.23 \pm 0.04$ (stat) ± 0.06 (syst) mag. The CCHP distance to M66 coincides with the distance modulus obtained with Cepheids by Saha et al. (1999) of $\mu = 30.22 \pm 0.12$ mag. The agreement between the distance obtained by Cepheids by Saha et al (1999) and the one quoted using TRGB by the CCHP is not a simple coincidence. The zero point adopted by Saha et al. (1999) was an LMC distance of $\mu = 18.5$ mag, while the CCHP adopted and LMC distance of $\mu = 18.48$ mag. This agreement has been found despite the fact of the crowding of Cepheids in M66 (the WFPC2 fields are quite crowded and of lower quality than modern observations taken with ACS or WFC3). Despite other possible systematic effects of the two methods, the zero point to LMC has been crucial in their agreement.

M66 looks like a good anchor for the BL SNe Ia in the Hubble flow. It is ideal for getting distances to SN 2011cn in IC 4758, SN 2008go in 2MASX J22104396-2047256, SN 1999ek in UGC 3329. Concerning the systematic error of the distance of the anchor, we assign an error of 0.04 mag to the distance of M66, as for the rest of the anchor galaxies.

For the comparison of SN 1989B and the SNe Ia which are BL but also high velocity (HV), we need to apply a shift to the reference SN 1989B in accordance with the HV they have. SNe Ia from the BL subtype move faster than those of the CN subtype. A shift in the spectral features to the blue to compare the spectra between BL subtypes is a step expected within our twin to twin

comparison. The amount for which one SN Ia of the BL subtype is moving faster than another one from the BL subtype is not a parameter in the Branch et al. (2006) classification.

4. SAMPLE OF SNE IA IN THE HUBBLE FLOW

The selection of twins has to follow the requirements of being within the same subtype of SNe Ia (within a given Branch et al. 2006 class), have a similar stretch within errors, and having various epochs for the spectra. A number of SNe Ia from the Carnegie Supernova Project (CSP) fulfil those criteria.

Table 2 shows the references for the spectra and photometry used in this paper. Some spectra come directly from the Carnegie Supernova Project archive, and some others taken from the *WISEREP*. All are carefully calibrated in flux in agreement with the published photometry. The error in the flux calibration is taken into account.

The comparison of the light curves of the Hubble flow SNe Ia with those nearby, is a first selection criterion in the use of the "SNe Ia twins for life" method. We have taken similar $\Delta m(B)_{15}$ and the stretch factor s_{BV}^D (see Tables 3 and 4). Burns et al. (2014) describe a relation between the SNooPy stretch and $\Delta m_{15}(B)$ as:

$$s_{BV} = 0.955 - 0.458(\Delta m_{15}(B) - 1.1) \quad (1)$$

where $\Delta m_{15}(B)$ is the decline rate parameter established by Phillips (1993), which is at the core of the discovery of the correlation of the brightness of a SN Ia and the rate of decline of the B light curve in the 15 days past maximum. The color stretch s_{BV} is the rate of decline of the B-V color in 30 days. One disadvantage of s_{BV} is that one needs to have a photometric B and V coverage from B maximum until approximately 40 days past maximum. The CSP and Burns et al. (2020) quote, instead of s_{BV} , the stretch s_{BV}^D , where the "D" in the superindex refers to the direct comparison in B, without having to take into account the *BVRI* light curves,

In this research, we have been very careful in selecting for the comparison of SNe Ia in the Hubble flow with their twins, SNe Ia with very similar $\Delta m_{15}(B)$ and stretch s_{BV}^D values.

4.1. The SN 2011fe-like in the Hubble flow

We have thus M101 and its supernova SN 2011fe at a distance of $d = 6.880 \pm 0.174$ Mpc ($\mu = 29.188 \pm 0.055$).

In the Hubble flow we have found an excellent twin in SN 2008bz. It has a $\Delta m(B)_{15}$ of 1.093 ± 0.079 . The $\Delta m(B)_{15}$ of SN 2011fe is 1.07 ± 0.006 .⁸ SN 2008bz's heliocentric redshift is 0.06, thus the full expansion of the modulus in q_0 is needed. The result can be seen in Figure 1. It is remarkable that with this method we can go from 6.8 Mpc to ~ 270 Mpc in a single step. We will come back to the result in distance in section 6.

Another SN Ia in the Hubble flow of the same type as SN 2011fe is ASASSN-15db in NGC 5996. It has a nearly identical light-curve decline value as SN 2011fe, with $\Delta m(B)_{15} = 1.089 \pm 0.061$ for ASASSN-15db and $\Delta m(B)_{15} = 1.07 \pm 0.006$ for SN 2011fe. Despite the fact that we do not have the full light curve of this SN Ia (which belongs to the DR of CSP II soon to come), we can calibrate the spectra with the photometry for the two phases.

4.2. SNe Ia in the Hubble flow like SN 2013aa/2017cbv

SNe Ia that can be considered twins of SN 2013aa are SN 2007A, SN 2008bf and LSQ14gov.

SN 2007A has $\Delta m(B)_{15} = 0.971 \pm 0.074$ which compares well with SN 2013aa and SN 2017cbv, since $\Delta m(B)_{15} = 0.95 \pm 0.01$ for SN 2013aa.⁹

This supernova has an internal reddening of 0.2 magnitudes. Such a high extinction by dust has been well captured by the SNooPY light curve fitter. It was not well captured by SALT2 and, therefore, the value of the distance derived by Pantheon+ to this supernova is too large.

SN 2008bf compares very well with SN 2013aa. The first one has a $\Delta m(B)_{15} = 0.97 \pm 0.06$ and the second one $\Delta m(B)_{15} = 0.95 \pm 0.01$. For the case of doubt about their similarity, a direct comparison of the light curves can be seen for this one and for all SNe Ia from CSP I (see Appendix A). SN 2008bf as a perfect twin of SN 2013aa.

For the last comparison, we have LSQ14gov which has $\Delta m(B)_{15} = 0.979 \pm 0.061$ to compare with $\Delta m(B)_{15} = 0.95 \pm 0.01$ of SN 2013aa.

⁸ The two SN 2011fe-like SNe Ia have the following stretches: SN 2008bz, $s_{BV}^D = 0.948 \pm 0.948 \pm 0.046$; ASASSN-15db, $s_{BV}^D = 0.953 \pm 0.04$. Both compare well with SN 2011fe, with $s_{BV}^D = 0.97 \pm 0.033$.

⁹ The stretches of the three SNe Ia of this group are: for SN 2007A, $s_{BV}^D = 1.025 \pm 0.061$; for SN 2008bf, $s_{BV}^D = 1.058 \pm 0.012$; for LSQ14gov, $s_{BV}^D = 1.141 \pm 0.041$. All of them compare well with the stretch of the anchor SN 2013aa: $s_{BV}^D = 1.11 \pm 0.02$.

Table 2. Spectra and Photometry used for the SN fits

Supernova	Galaxy	z_{hel}	Spectral range (Å)	Phase	Date	Reference
SN 2012bo	NGC 4726	0.025431	3665-8847	+14	2012-04-18	WiSeREP
...	...	...	3634-9602	+26	2012-04-30	...
SN 2008bq	ESO 308-G25	0.03400	3675-8931	+5.57	2008-04-12	CSP-DR1
...	...	...	...	+33.44	2008-05-10	...
SN 2008bz	A123857+1107	0.0603	3626-9433	+4.65	2008-04-26	...
...	...	...	3800-9235	+13	2008-05-05	...
LSQ12fxd	ESO 487-G004	0.0312	3352-7475	-2	2012-11-13	CSP-DR1
...	...	...	3579-9615	+3.7	2012-11-19	...
SN 2008bf	NGC 4055/4057	0.0235	3715-9974	-2.70	2008-03-26	...
...	...	...	3710-9018	+42.07	2008-05-11	...
SN 2007A	NGC 105	0.01764	3473-7409	-2.86	2007-01-10	WiSeREP
...	...	...	3378-1027	+13.14	2007-02-24	...
ASASSN-15db	NGC 5996	0.01099	3604-9164	+3	2015-02-25	...
...	...	...	...	+4	2015-02-26	...
LSQ14gov	GALEXMSC J040601.67-160139.7	0.0896	3650-9250	-6	2013-12-30	...
...	...	...	3430-9100	+1	2014-01-06	...
SN 2007ca	MCG -02-34-61	0.01407	3455-9494	+1.4	2007-05-08	CSP-DR1
...	...	...	3297-9416	+4.2	2007-05-11	...
SN 2001cn	IC 4758	0.0156	3827-7218	+2.2	2001-06-14	...
...	...	...	3715-7115	+10	2001-06-22	...
SN 2008go	2MASX J22104396-2047256	0.0623	3800-9234	+0.6	2008-10-27	...
...	...	...	3636-9450	+7.5	2008-11-04	...
SN 1999ek	UGC 3329	0.0177	3720-7540	+3.3	1999-11-03	...
...	...	...	3720-7540	+9.32	1999-11-09	...
Photometry						
SN 2012bo	...	...	...	...	Morrell	...
SN 2008bq	...	...	...	...	CSP-DR1	...
SN 2008bz	...	...	...	...	CSP-DR1	...
LSQ12fxd	...	...	...	...	Morrell	...
SN 2008bf	...	...	...	...	CSP-DR1	...
SN 2007A	...	...	...	...	CSP-DR1	...
ASASSN-15db	...	...	...	...	Morrell	...
LSQ14gov	...	...	...	...	Morrell	...
SN 2007ca	...	...	...	...	CSP-DR1	...
SN 2001cn	...	...	...	...	Krisciunas	...
SN 2008go	...	...	...	...	CSP-DR1	...
SN 1999ek	...	...	...	...	Krisciunas	...

4.3. SNe Ia in the Hubble flow like SN 2013dy

The advantage of these SNe Ia is that there is a pretty good agreement between the distances given by the two groups for this anchor.

Four SNe Ia from the SN 2013dy class are SN 2012bo and SN 2008bq, SN 2007ca and LSQ12fxd.

SN 2012bo has a $\Delta m(B)_{15} = 0.931 \pm 0.061$ which compares well with the $\Delta m(B)_{15} = 0.92 \pm 0.006$ from SN 2013dy.¹⁰

¹⁰ The stretches of the four SNe Ia of this group are: for SN 2012bo, $s_{BV}^D = 1.160 \pm 0.0411$; for SN 2008bq, $s_{BV}^D = 1.056 \pm 0.014$; for SN 2007ca, $s_{BV}^D = 1.077 \pm 0.012$; for LSQ12fxd, $s_{BV}^D = 1.161 \pm 0.040$. All of them compare well with the stretch of the anchor SN 2013dy: $s_{BV}^D = 1.091 \pm 0.03$.

We have paired SN 2008bq with SN 2013dy. It has a similar $\Delta m(B)_{15}$ of 0.90 ± 0.062 versus 0.92 ± 0.006 for SN 2013dy.

SN 2007ca has $\Delta m(B)_{15} = 0.904 \pm 0.032$ while $\Delta m(B)_{15}$ is 0.92 ± 0.006 for SN 2013dy. They are very good twins.

LSQ12fxd has $\Delta m(B)_{15} = 0.940 \pm 0.06$, versus $\Delta m(B)_{15} = 0.92 \pm 0.006$ of SN 2013dy.

4.4. SNe Ia in the Hubble flow like SN 1989B

SN 1989B is a broadline SNe Ia that appeared in M66, a nearby galaxy for which there is a coincident distance by TRGBs (Hoyt et al. 2019; Freedman et al. 2019) and Cepheids (Saha et al. 1999). There is no published distance by SH0ES because SN 1989B shows too much host reddening to be in the fiducial SH0ES samples.

Three SNe Ia in the CSP I are broad line and have a rate of decline in the light curve similar to SN 1989B. Usually broadline supernovae have also high velocity ejecta.

Table 3. SNe 2012bo, 2008bq, 2008bz, LSQ12fxd, 2008bf, 2007A

SN 2012bo in NGC 4726		
RA, DEC ^a	12:50:45.23	-14:16:08.5
Discovery date ^a	...	2012-03-27
Phase (referred to maximum light) ^b	...	-8 days
Redshift ^b	...	0.026548
E(B-V) _{MW} ^c	...	0.046 mag
Stretch factor s_{BV}^D ^b	...	1.160±0.0411
$\Delta m(B)_{15}$ ^b	...	0.931±0.061
Phases of the spectra used	...	14, 26 days
^a Itagaki et al. (2012). ^b Morrell et al. (2023).		
^c Schlaflly & Finkbeiner (2011).		
SN 2008bq in ESO 308-G25		
RA, DEC ^a	06:41:02.51	-38:02:19.0
Discovery date ^a	...	2008-04-02
Phase (referred to maximum light) ^b	...	-4 days
Redshift ^b	...	0.03446
E(B-V) _{MW} ^c	...	0.080 mag
Stretch factor s_{BV}^D ^d	...	1.056±0.014
$\Delta m(B)_{15}$ ^d	...	0.895±0.062
Phases of the spectra used	...	5.57, 33.44 days
^a Luckas et al. (2008). ^b Folatelli et al. (2013).		
^c Schlaflly & Finkbeiner (2011). ^d Krisciunas et al. (2017).		
SN 2008bz in A123957+1107		
RA, DEC ^a	12:38:57.74	+11:07:46.2
Discovery date ^a	...	2008-04-22
Phase (referred to maximum light) ^b	...	≈0 days
Redshift ^b	...	0.0614
E(B-V) _{MW} ^c	...	0.023 mag
Stretch factor s_{BV}^D ^d	...	0.948±0.046
$\Delta m(B)_{15}$ ^b	...	1.093±0.079
Phases of the spectra used	...	4.65, 13 days
^a Yuan et al. (2008). ^b Krisciunas et al. (2020).		
^c Schlaflly & Finkbeiner (2011). ^d Morrell et al. (2023).		
LSQ12fxd in ESO487-G004		
RA, DEC ^a	05:22:17.02	-25:35:47.1
Discovery date ^a	...	2012-10-31
Phase (referred to maximum light) ^b	...	-15 days
Redshift ^b	...	0.031
E(B-V) _{MW} ^c	...	0.022 mag
Stretch factor s_{BV}^D ^b	...	1.161±0.040
$\Delta m(B)_{15}$ ^b	...	0.940±0.060
Phases of the spectra used	...	-2, 3.7 days
^a Bufano et al. (2012). ^b Morrell et al. (2023).		
^c Schlaflly & Finkbeiner (2011).		
SN 2008bf in NGC 4055/4057		
RA, DEC ^a	12:04:02.90	+20:14:42.6
Discovery date ^a	...	2008-03-18
Phase (referred to maximum light) ^b	...	-11 days
Redshift ^b	...	0.02510
E(B-V) _{MW} ^c	...	0.036 mag
Stretch factor s_{BV}^D ^d	...	1.058±0.012
$\Delta m(B)_{15}$ ^d	...	0.97±0.060
Phases of the spectra used	...	-2.70, 42.07 days
^a Panski et al.(2008). ^b Folatelli et al. (2013).		
^c Schlaflly & Finkbeiner (2011). ^d Krisciunas et al. (2017).		
SN 2007A in NGC 105		
RA, DEC ^a	00:25:16.66	+12:53:12.5
Discovery date ^a	...	2007-01-02
Phase (referred to maximum light) ^b	...	-10 days
Redshift ^b	...	0.01648
E(B-V) _{MW} ^c	...	0.074 mag
Stretch factor s_{BV}^D ^d	...	1.012±0.061
$\Delta m(B)_{15}$ ^d	...	0.971±0.074
Phases of the spectra used	...	-2.86, 13.14 days
^a Picket et al. (2007). ^b Folatelli et al. (2013).		
^c Schlaflly & Finkbeiner (2011). ^d Krisciunas et al. (2017).		

Table 4. SNe ASASSN-15db, LSQ14gov, 2007ca, 2001cn, 2008go, 1999ek

ASASSN-15db in NGC 5996		
RA, DEC ^a	15:46:58.9	17:53:03
Discovery date ^a	...	2015-02-15
Phase (referred to maximum light) ^b	...	-7 days
Redshift ^b	...	0.0114
E(B-V) _{MW} ^c	...	0.03 mag
Stretch factor s_{BV}^D ^b	...	0.953±0.040
$\Delta m(B)_{15}$ ^b	...	1.089±0.061
Phases of the spectra used	...	3, 4 days
^a Holoien et al. (2015). ^b Morrell et al. (2023).		
^c Schlaflly & Finkbeiner (2011).		
LSQ14gov in GALEXMSC J040601.67-160149.7		
RA, DEC ^a	04:06:01.53	-16:01:38.86
Discovery date ^a	...	2014-12-21
Phase (referred to maximum light) ^b	...	-15 days
Redshift ^b	...	0.0896
E(B-V) _{MW} ^c	...	0.0377 mag
Stretch factor s_{BV}^D ^b	...	1.141±0.041
$\Delta m(B)_{15}$ ^b	...	0.979±0.061
Phases of the spectra used	...	-6, 1 days
^a Ducrof et al. (2014). ^b Morrell et al. (2023).		
^c Schlaflly & Finkbeiner (2011).		
SN 2007ca in MGC-02-34-61		
RA, DEC ^a	13:31:05.81	-15:06:06.6
Discovery date ^a	...	2007-04-25
Phase (referred to maximum light) ^b	...	-12 days
Redshift ^b	...	0.01508
E(B-V) _{MW} ^b	...	0.067
Stretch factor s_{BV}^b ^b	...	1.077±0.012
$\Delta m(B)_{15}$ ^b	...	0.904±0.032
Phases of the spectra used	...	1.4, 4.2 days
^a Itagaki et al. (2007). ^b Uddin et al. (2020).		
SN 2001cn in IC 4758		
RA, DEC ^a	18:46:17.84	-65:45:41.8
Discovery date ^a	...	2001-06-12
Phase (referred to maximum light) ^b	...	-4 days
Redshift ^b	...	0.0156
E(B-V) _{MW} ^c	...	0.052
Stretch factor s_{BV}^b ^b	...	0.927±0.020
$\Delta m(B)_{15}$ ^b	...	1.080±0.044
Phases of the spectra used	...	2.2, 10 days
^a Chassagne (2001) ^b Morrell et al. (2024)		
^c Schlaflly & Finkbeiner (2011).		
SN 2008go in 2MASX J22104396-2047256		
RA, DEC ^a	22:10:44.03	-20:47:17.2
Discovery date ^a	...	2008-10-22
Phase (referred to maximum light) ^b	...	-5 days
Redshift ^c	...	0.06
E(B-V) _{MW} ^b	...	0.033
Stretch factor s_{BV}^b ^b	...	0.939±0.044
$\Delta m(B)_{15}$ ^b	...	1.093±0.089
Phases of the spectra used	...	0.6, 7.5 days
^a Griffith et al. (2008). ^b Stahl et al. (2019).		
^c Schlaflly & Finkbeiner (2011).		
SN 1999ek in UGC 3329		
RA, DEC ^a	05:36:31.60	+16:38:17.8
Discovery date ^a	...	1999-10-20
Phase (referred to maximum light) ^b	...	-11 days
Redshift ^b	...	0.0177
E(B-V) _{MW} ^b	...	0.502
Stretch factor s_{BV}^b ^b	...	0.921±0.008
$\Delta m(B)_{15}$ ^b	...	1.097±0.019
Phases of the spectra used	...	3.3, 9.32 days
^a Johnson & Li (1999). ^b Morrell et al. (2024).		

This produces a blueshift in the lines of the supernova. SN 1989B is not a high velocity supernova. Two of the broadline SNe Ia at high- z have high velocity features: SN 2001cn and SN 2008go. In contrast, SN 1999ek is a broad line that does not exhibit high velocity in the its spectral features.

We have paired SN 2001cn, SN 2008go and SN 1999ek with SN 1989B. There will be a blueshift applied to SN 1989B to compare with SN 2001cn and SN 2008go, whereas in SN 1999ek it is not necessary.

SN 2001cn has $\Delta m(B)_{15} = 1.080 \pm 0.044$, which fits well with $\Delta m(B)_{15} = 1.053 \pm 0.11$ of SN 1989B.

SN 2008go has $\Delta m(B)_{15} = 1.093 \pm 0.089$, also concordant with the $\Delta m(B)_{15} = 1.053 \pm 0.11$ of SN 1989B.

SN 1999ek has $\Delta m(B)_{15} = 1.097 \pm 0.019$, again compatible with the $\Delta m(B)_{15} = 1.053 \pm 0.11$ of the anchor¹¹

5. METHODOLOGY

We have used eight SNe Ia from Carnegie I included in the DR1. Their characteristics are described in Folatelli et al. (2012) and four SNe Ia in Carnegie II (Morrell et al. 2024). We have been able to calibrate the spectra, which are uploaded to the *WISeREP*.

All the spectra have been corrected from Galactic reddening. This is done with the purpose of allowing to compare Galactic dust-free spectra. It also helps to learn about dust extinction in the host. The spectra are compared in the frame of the Hubble flow supernova. For that purpose nearby spectra are shifted in wavelength and their flux is diluted in accordance with the $(1+z)$ law. The shift brings the nearby SN Ia to the redshift of the Hubble flow supernova taking into account the heliocentric redshift of the nearby SNIa (shifting by $(1+z_{flow})/(1+z_{nearby})$, though the redshift of nearby SNe Ia is always low).

The method primarily determines the distance between the nearby SN Ia and that in the Hubble flow. It will also give the relative $\Delta E(B-V)$ reddening. The $\Delta E(B-V)$ value will indicate whether $E(B-V)$ is larger or smaller for one of the two SNe with respect to its twin, taken as

¹¹ Concerning the stretch, SN 1989B has $s_{BV}^D = 0.954 \pm 0.034$. The corresponding values for its distant twins are: for SN 2001cn, $s_{BV}^D = 0.927 \pm 0.020$; for SN 2008go, $s_{BV}^D = 0.939 \pm 0.044$; for SN 1999ek, $s_{BV}^D = 0.921 \pm 0.008$. So, the stretches of the three distant SNe Ia compare well with that of the anchor SN 1989B.

reference. Concerning the distance modulus, in reality we are calculating distances relative to the SNe Ia of reference. So, we have a distance scale factor between the reference and the SN in the Hubble flow. For the sake of producing a final result, we give the absolute distance modulus μ .

To find the best values and the uncertainties of all variables, we explore the parameter space with Markov Chain Monte Carlo (MCMC) techniques after converting the χ^2 into a log likelihood function (the probability of a dataset given the model parameters) that we aim to optimize:

$$\log \mathcal{L} = -0.5 * \sum (f_{\lambda}^{obs} - f_{\lambda}^{ref})^2 / \sigma_{\lambda}^2 + \log \sigma_{\lambda} \quad (2)$$

where *obs* corresponds to a given supernova and *ref* to its reference twin, which represents a whole class. f_{λ}^{obs} and f_{λ}^{ref} correspond to the fluxes of the two SNe at different wavelengths, and σ_{λ} , defined as $\sigma_{\lambda}^2 = \epsilon_{obs}^2 + \epsilon_{ref}^2 + \epsilon_{int}^2$ includes the quadratic addition of the uncertainty on the fluxes of both SNe and the intrinsic uncertainty in the flux comparison. This uncertainty can be described as the maximum flux of the observed SNIa multiplied by a factor, η_{obs} , as $\epsilon_{int} = \max(f_{\lambda}^{obs}) * \eta_{obs}$. This intrinsic uncertainty absorbs the remaining uncertainty to help to maximize the likelihood. We note that in RLGH24, this intrinsic uncertainty was already included with a slightly different definition using the f_{λ}^{obs} multiply by the factor η_{obs} . This new definition of ϵ_{int} provides an estimate of the intrinsic magnitude scatter, as $\sigma_{int} = (2.5/\ln(10)) \times (\epsilon_{int}/\max(f_{\lambda}^{obs}))$.

The MCMC allows to get the maximum likelihood values of each parameter from the posterior distribution of samples, together with their uncertainties, including the propagation of all the parameter uncertainties on each parameter.

In fact, as we are using different phases, we have a total likelihood function. For two phases, n corresponds to 1 and 2 below. Thus the total likelihood function is written as

$$\log \mathcal{L}_n = \log \mathcal{L}_1 + \log \mathcal{L}_2 \quad (3)$$

i.e. just the addition of all individual SN phases.

We use a Python package, EMCEE (Foreman-Mackey et al. 2013)¹² to explore the likelihood of each variable. EMCEE utilizes an affine invariant MCMC ensemble sampler proposed by Goodman et al. (2010).

¹² <https://emcee.readthedocs.io/>

This sampler tunes only two parameters to get the desired output: number of walkers and number of steps. The run starts by assigning initial maximum likelihood values of the variables to the walkers. The walkers then start wandering and explore the full posterior distribution. After an initial run, we inspect the samplers for their performance. We do this by looking at the time series of variables in the chain and computing the autocorrelation time, τ^{13} . In our case, the maximum autocorrelation time among the different variables is about 50. When the chains are sufficiently burnt-in, they forget their initial start point), we can safely throw away some steps that are a few times higher than the burnt-in steps. In our case, we run EMCEE with 32 walkers and 10,000 steps and throw away the first ~ 250 samples, equivalent to ~ 4 times the maximum autocorrelation time. Thus, our burn-in value in this computation is equal to 4 times the maximum autocorrelation time. A criterion of good sampling is the acceptance fraction, a_f . This is the fraction of steps that are accepted after the sampling is done. The suggested value of a_f is between 0.2 - 0.5 (Gelman et al. 1996). In each run, we typically obtain $a_f \sim 0.25$.

The priors in the difference between $E(B-V)$ of the host galaxies go from $[-0.4, +0.4]$. As we said before, we correct from Galactic reddening the SNe Ia spectra before running the MCMC. So our intrinsic $\Delta(E(B-V))$ is entirely attributed to the host galaxies. In relation to the priors in distance for each supernova, they vary depending of the particular SN Ia.¹⁴

One can visualize the output of two-dimensional and one-dimensional posterior probability distributions in a corner plot corresponding to 1σ , 2σ , 3σ . This corner plot shows one and two dimensional projections of the posterior probability distributions of our parameters. We present the results of the MCMC for all the SNe Ia in the Hubble flow that have been included in this paper. We give the results for each phase and the combined phase (see Figures 1 to 12).

¹³ <https://emcee.readthedocs.io/en/stable/tutorials/autocorr/>

¹⁴ Distance priors are (in Mpc): [95.0, 115.0] (SN 2012bo); [130.0, 155.0] (SN 2008bq); [200.0, 350.0] (SN 2008bz); [110.0, 140.0] (LSQ12fxd); [101.0, 110.0] (SN 2008bf); [62.0, 82.0] (SN 2007A); [35.0, 65.0] (ASASSN-15db); [370.0, 410.0] (LSQ14gov); [40.0, 80.0] (SN 2007ca); [50.0, 80.0] (SN 2001cn); [230.0, 300.0] (SN 2008go); [60.0, 80.0] (SN 1999ek).

6. RESULTS AND DISCUSSION

6.1. The test of the three rung method

The present work can serve to test the three rung method for the determination of H_0 . In this respect, the results obtained by the SH0ES-Pantheon+ collaboration and by our method using only one step are the same. That could in principle not have been the case, as the 3 rung method goes through the calibration of a fiducial absolute magnitude of SNe Ia which is not required by our method.

We include in Table 5 the distance moduli using the "SNe Ia twins for life" method as well as those provided by Pantheon+ with SH0ES and by the Carnegie Supernova Project. Those last two methods are calibrated with the Cepheid distances. The SNe Ia moduli in the Hubble flow from the Pantheon+ and SH0ES are quite close to those from the CCHP.

The Pantheon+ data come with their errors in distance moduli in their release and the CCHP distance moduli and errors are generated as in Uddin et al. (2024) code given the entries in the light curve parameterization. We see similar distances moduli in Table 5 by our method when compared with those from Pantheon+ and with Cepheids by the CCHP. There are some differences (in 3 SNe Ia) mostly due to the underestimate of reddening (see Figure 13), as we will discuss in the next subsection.

To obtain the error in the "SNe Ia twins" distance modulus we add the various uncertainty terms. We take a systematic error due to anchors to be of 0.04 mag.

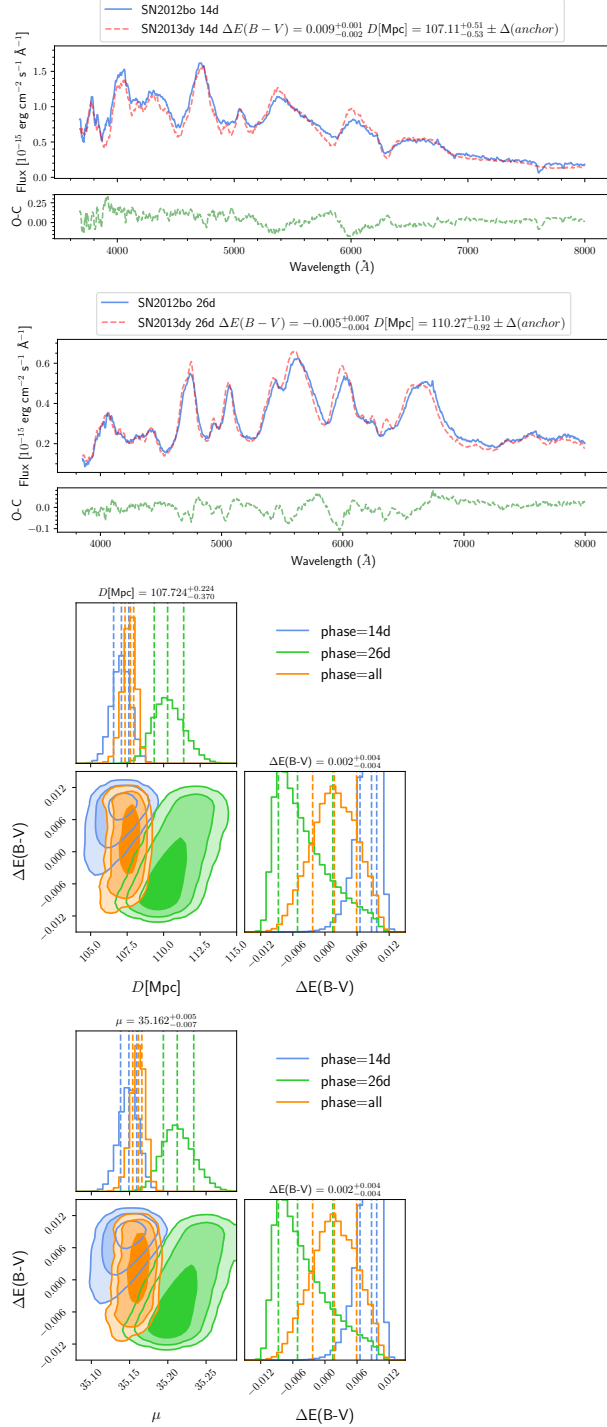


Figure 1. Top: Comparison of the spectrum at 14 days past maximum of SN 2012bo with that of SN 2013dy at the same phase. Middle below top: Comparison of the spectrum of SN 2012bo at 26 days past maximum with that of SN 2013dy at the same phase. Middle above bottom: Corner plot with the posterior probability at 1σ , 2σ , 3σ of distance and relative intrinsic reddening of SN 2012bo in relation to SN 2013dy. Bottom: Corner plot with the posterior probability at 1σ , 2σ , 3σ of distance moduli and relative intrinsic reddening of SN 2012bo in relation to SN 2013dy.

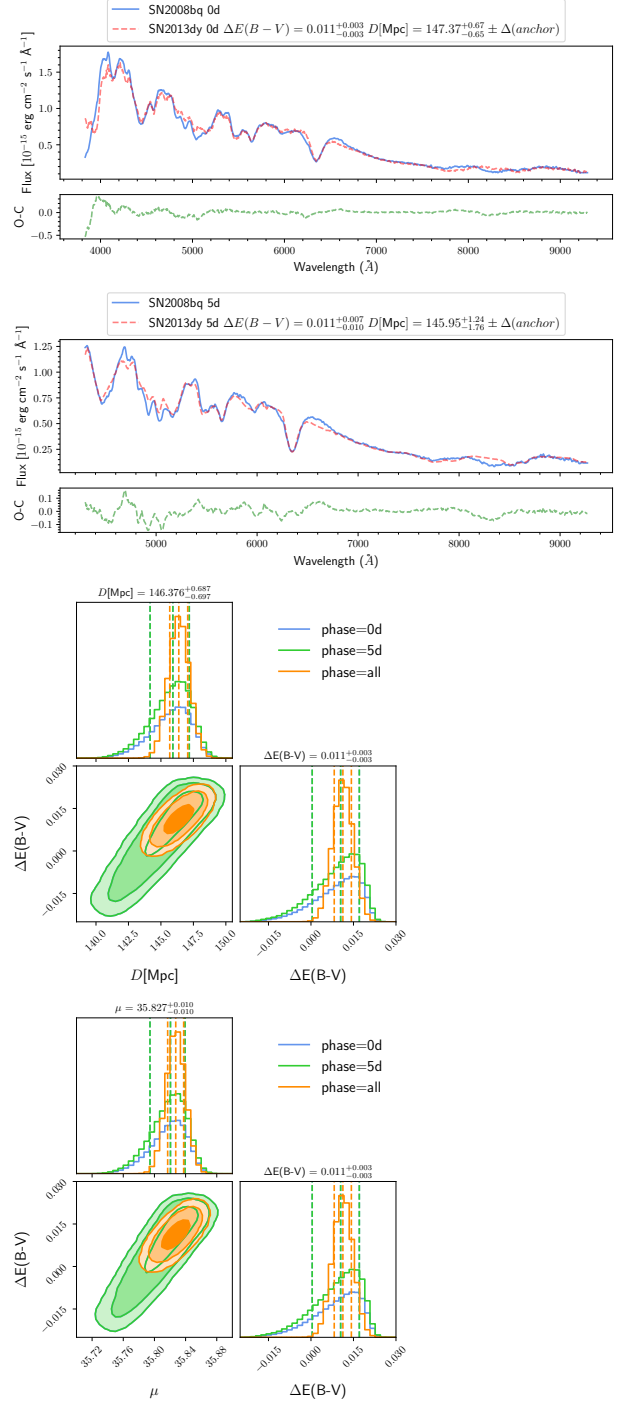


Figure 2. Top: Comparison of the spectrum at 0 days past maximum of SN 2008bq with that of SN 2013dy at the same phase. Middle below top: Comparison of the spectrum at 5 days past maximum with that of SN 2013dy at the same phase. Middle above bottom: Corner plot with the posterior probability at 1σ , 2σ , 3σ of distance and relative intrinsic reddening of SN 2008bq in relation to SN 2013dy. Bottom: Corner plot with the posterior probability at 1σ , 2σ , 3σ of distance moduli and relative intrinsic reddening of SN 2008bq in relation to SN 2013dy.

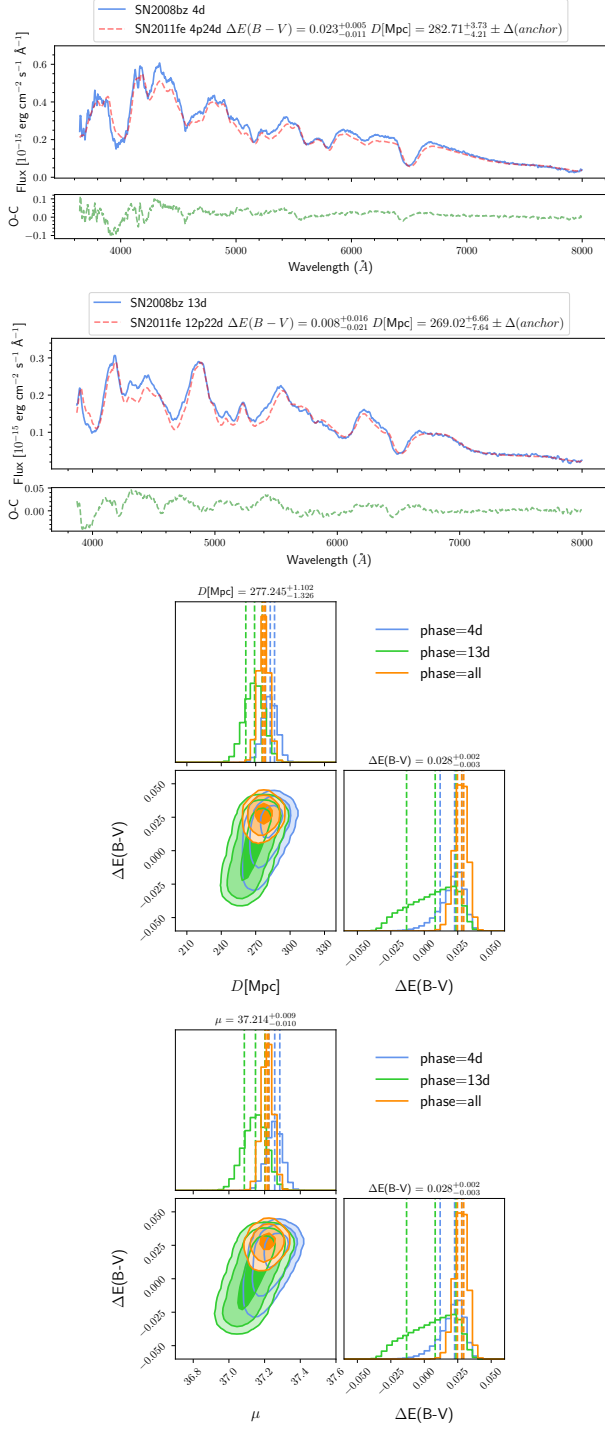


Figure 3. Top: Comparison of the early time spectrum of SN 2008bz with SN 2011fe at 4 days past maximum light. Middle below top: Comparison of the spectrum of SN 2008bz at 13 days past maximum with the corresponding spectrum of SN 2011fe. Middle above bottom: Corner plot with the posterior probability at 1σ , 2σ , 3σ of distance and relative intrinsic reddening of SN 2008bz in relation to SN 2011fe. Bottom: Corner plot with the posterior probability at 1σ , 2σ , 3σ of distance moduli and relative intrinsic reddening of SN 2008bz in relation to SN 2011fe.

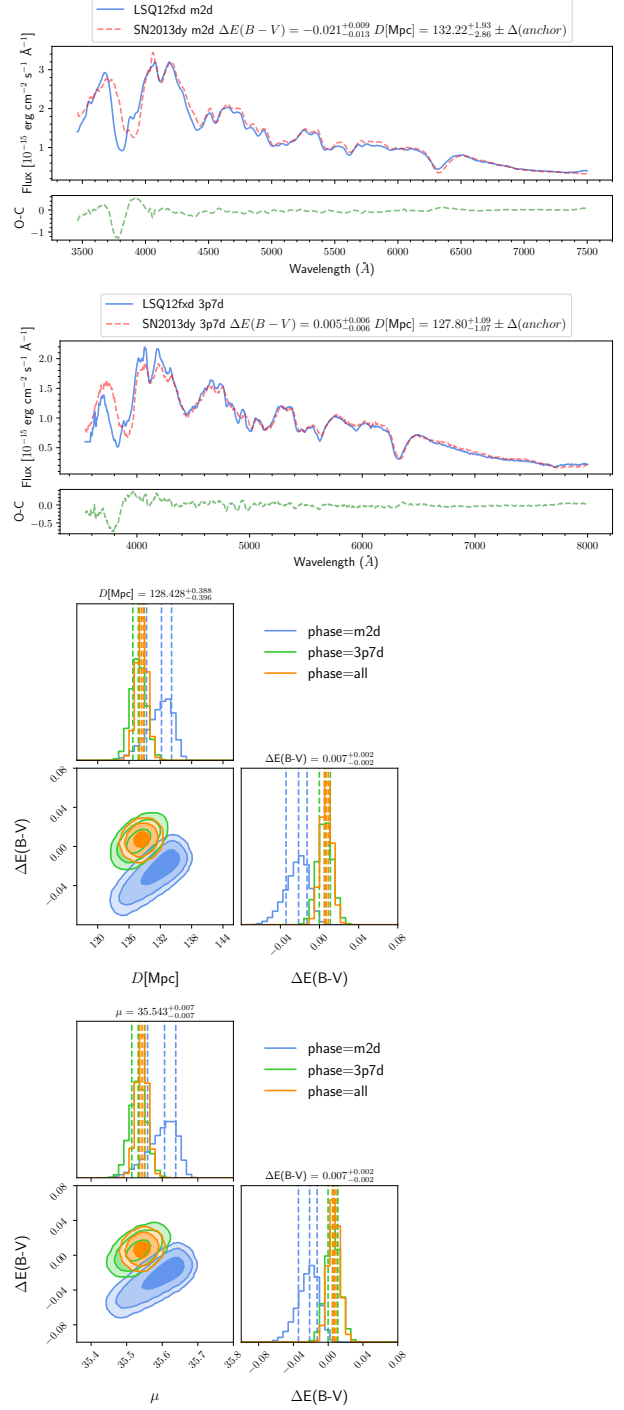


Figure 4. Top: Comparison of the spectrum of LSQ12fxd at 2 days before maximum with that of SN 2013dy in the same phase. Middle below top: Comparison of the spectrum of LSQ12fxd at 3.7 days past maximum with that of SN 2013dy. Middle above bottom: Corner plot with the posterior probability at 1σ , 2σ , 3σ of distance and relative intrinsic reddening of LSQ12fxd in relation to SN 2013dy. Bottom: Corner plot with the posterior probability at 1σ , 2σ , 3σ of distance moduli and relative intrinsic reddening of LSQ12fxd in relation to SN 2013dy.

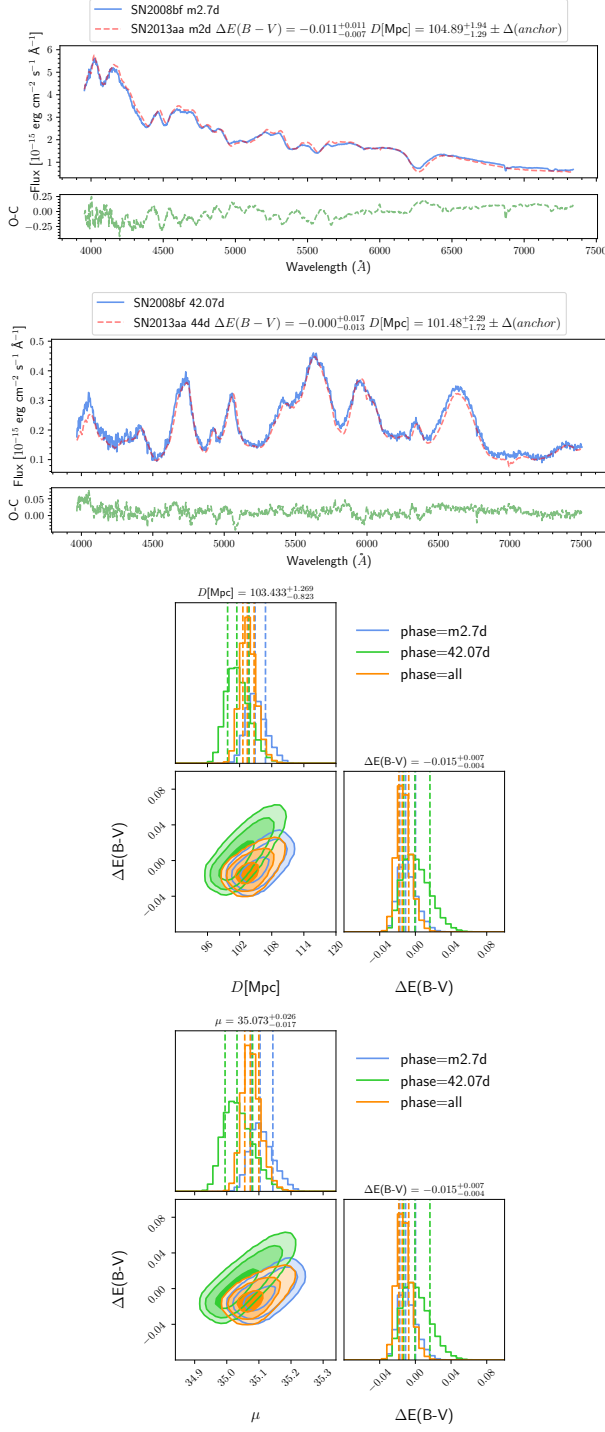


Figure 5. Top: Comparison of the early time spectrum of SN 2008bf with SN 2013aa at -2 days before maximum light. Middle below top: Comparison of the spectrum of SN 2008bf at 42 days past maximum with the corresponding spectrum of SN 2003aa. Middle above bottom: Corner plot with the posterior probability at 1σ , 2σ , 3σ of distance and relative intrinsic reddening of SN 2008bf in relation to SN 2013aa. Bottom: Corner plot with the posterior probability at 1σ , 2σ , 3σ of distance moduli and relative intrinsic reddening of SN 2008bf in relation to SN 2013aa.

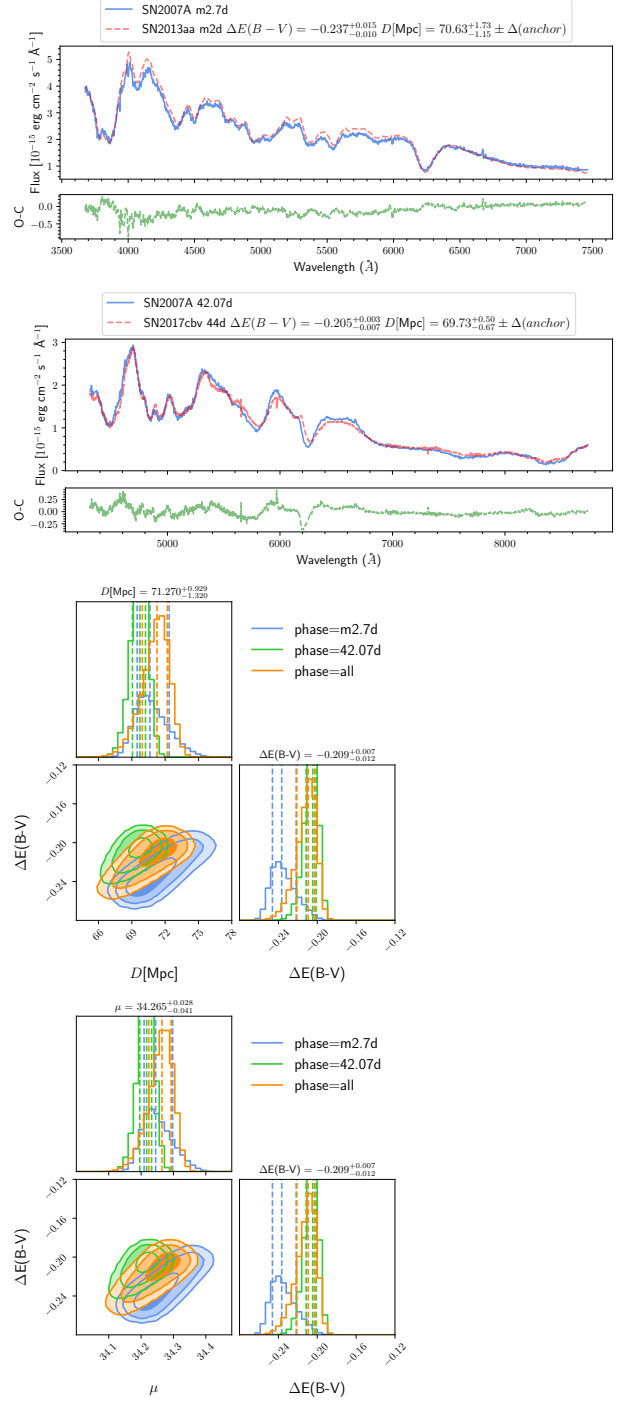


Figure 6. Top: Comparison of the spectrum at 2 days before maximum of SN 2007A with that of SN 2013aa at the same phase. Middle below top: Comparison of the spectrum at 42 days past maximum of SN 2007A with that of SN 2013aa at the same phase. Middle above bottom: Corner plot with the posterior probability at 1σ , 2σ , 3σ of distance and relative intrinsic reddening of SN 2007A in relation to SN 2013aa. Bottom: Corner plot with the posterior probability at 1σ , 2σ , 3σ of distance moduli and relative intrinsic reddening of SN 2007A in relation to SN 2013aa.

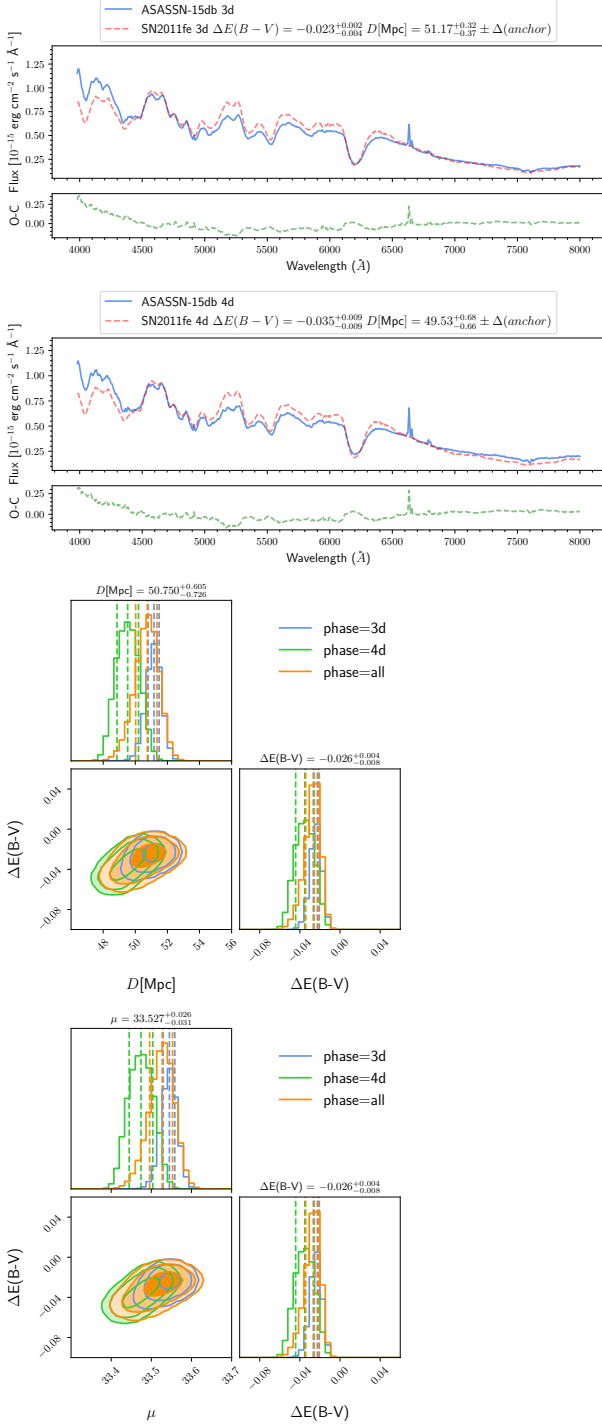


Figure 7. Top: Comparison of the spectrum at 3 days past maximum of ASASSN-15db with that of SN 2011fe at the same phase. Middle below top: Comparison of the spectrum of ASASSN-15db at 4 days past maximum with that of SN 2011fe at the same phase. Middle above bottom: Corner plot with the posterior probability at 1σ , 2σ , 3σ of distance and relative intrinsic reddening of ASASSN-15db in relation to SN 2011fe. Bottom: Corner plot with the posterior probability at 1σ , 2σ , 3σ of distance moduli and relative intrinsic reddening of ASASSN-15db in relation to SN 2011fe.

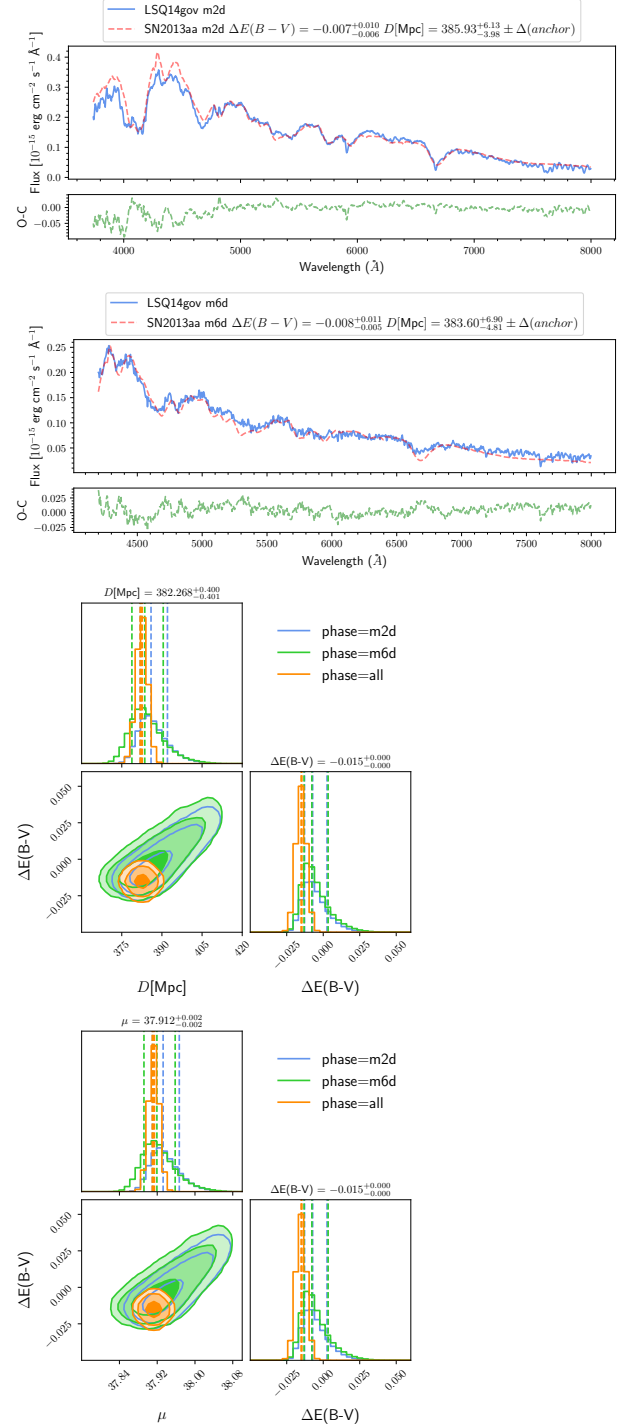


Figure 8. Top: Comparison of the spectrum at 1 day past maximum of LSQ14gov with that of SN 2013aa at the same phase. Middle below top: Comparison of the spectrum at 6 days before maximum with that of SN 2013aa at a similar phase. Middle above bottom: Corner plot with the posterior probability at 1σ , 2σ , 3σ of distance and relative intrinsic reddening of LSQ14gov in relation to SN 2013aa. Bottom: Corner plot with the posterior probability at 1σ , 2σ , 3σ of distance moduli and relative intrinsic reddening of LSQ14gov in relation to SN 2013aa.

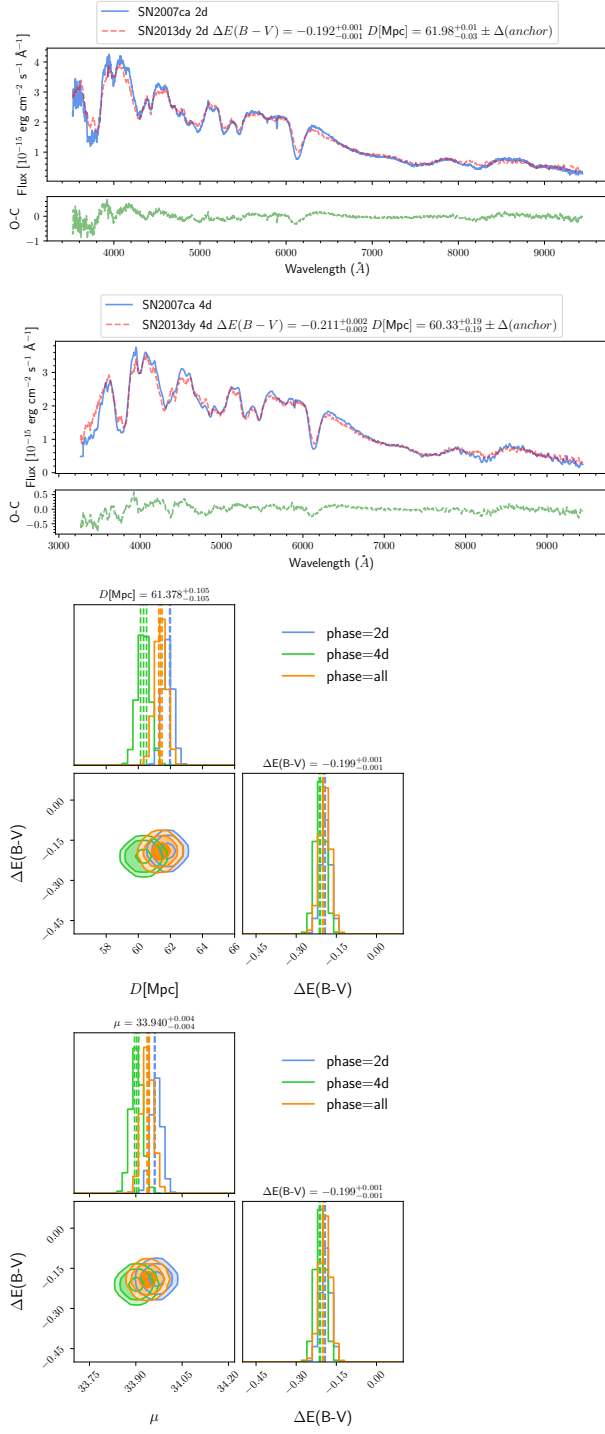


Figure 9. Top: Comparison of the spectrum at 2 days past maximum of SN2007ca with that of SN 2013dy at the same phase. Middle below top: Comparison of the spectrum at 4 days past maximum with that of SN 2013aa at a similar phase. Middle above bottom: Corner plot with the posterior probability at 1 σ , 2 σ , 3 σ of distance and relative intrinsic reddening of SN2007ca in relation to SN 2013dy. Bottom: Corner plot with the posterior probability at 1 σ , 2 σ , 3 σ of distance moduli and relative intrinsic reddening of SN2007ca in relation to SN 2013dy.

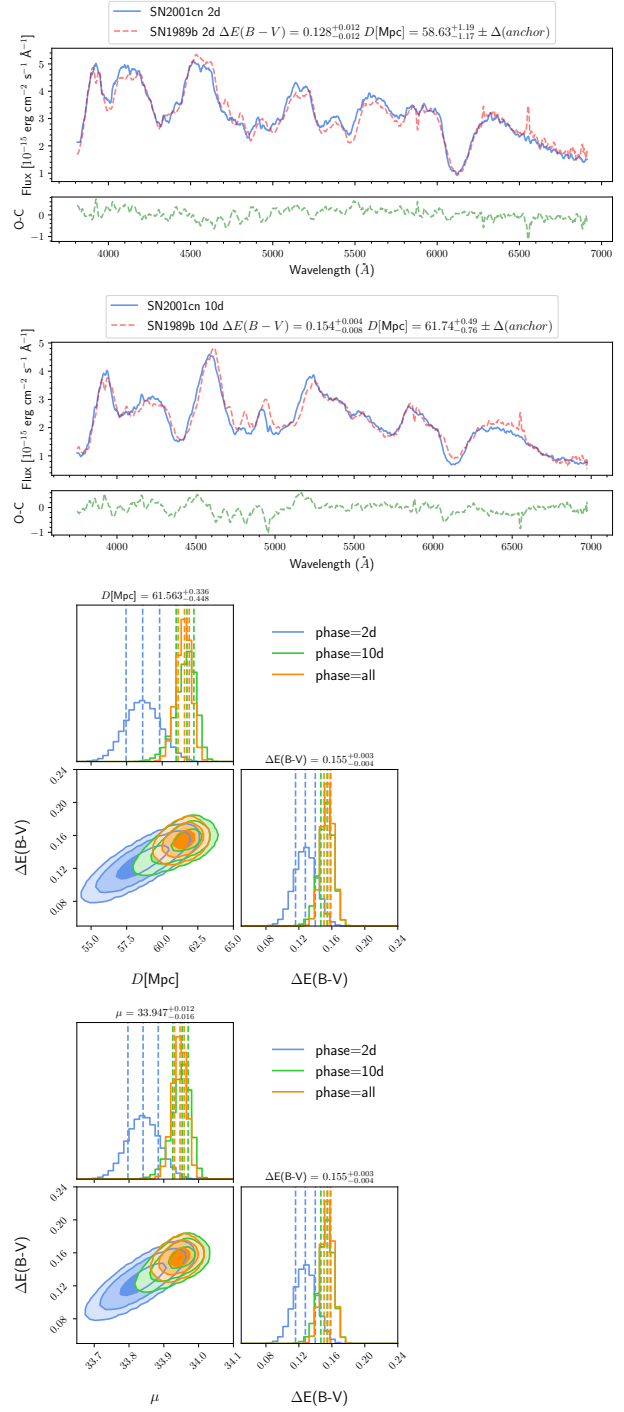


Figure 10. Top: Comparison of the spectrum at 2 days past maximum of SN 2001cn with that of SN 1989B at the same phase. Middle below top: Comparison of the spectrum at 10 days past maximum of SN 2001cn with that of SN 1989B at about the same time. Middle above bottom: Corner plot with the posterior probability at 1 σ , 2 σ , 3 σ of distance and relative intrinsic reddening of SN 2001cn in relation to SN 1989B. Bottom: Corner plot with the posterior probability at 1 σ , 2 σ , 3 σ of distance moduli and relative intrinsic reddening of SN 2001cn in relation to SN 1989B.

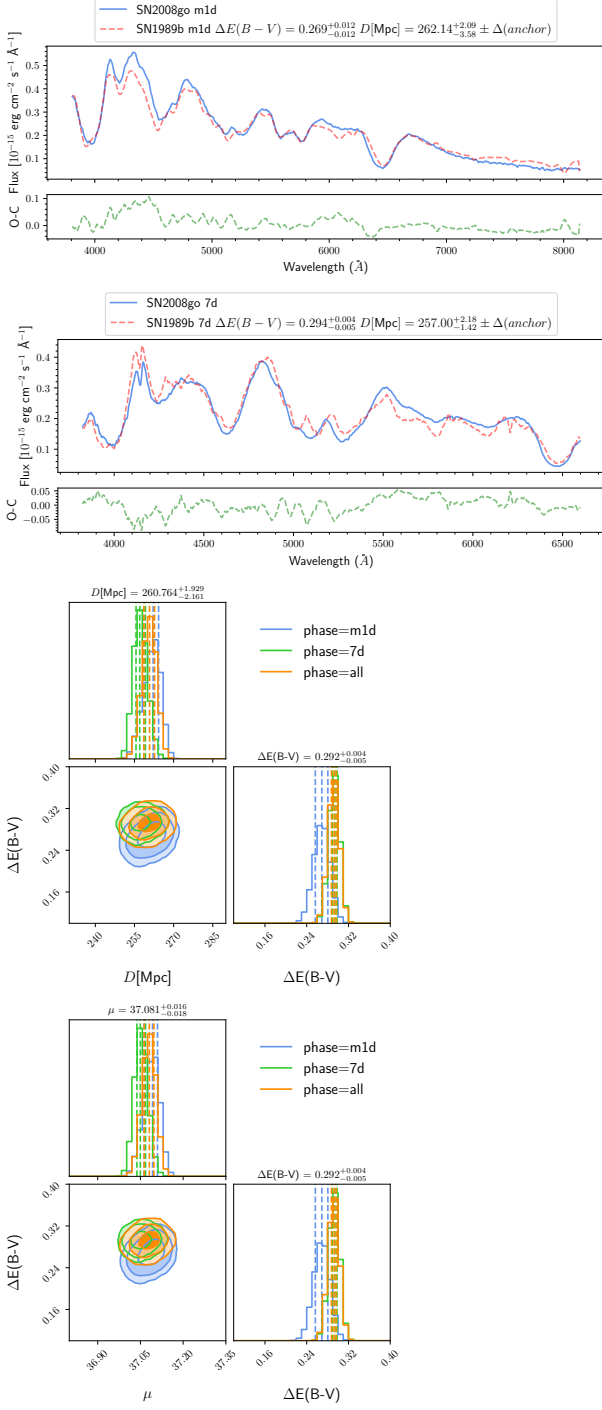


Figure 11. Top: Comparison of the spectrum at 1 day before maximum of SN 2008go with that of SN 1989B at the same phase. Middle below top: Comparison of the spectrum at 7 days past maximum of SN 2008go with that of SN 1989B at about the same phase. Middle above bottom: Corner plot with the posterior probability at 1 σ , 2 σ , 3 σ of distance and relative intrinsic reddening of SN 2008go relation to SN 1989B. Bottom: Corner plot with the posterior probability at 1 σ , 2 σ , 3 σ of distance moduli and relative intrinsic reddening of SN 2008go in relation to SN 1989B.

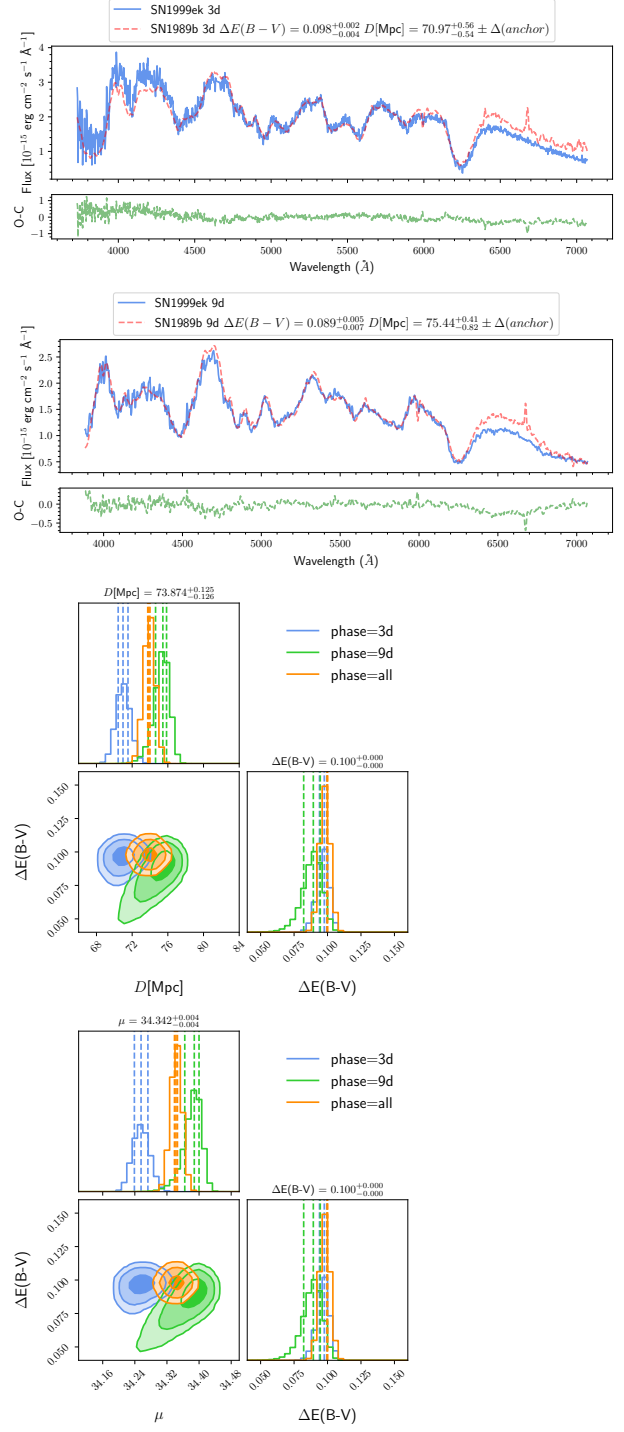


Figure 12. Top: Comparison of the spectrum at 3 days past maximum of SN 1999ek with that of SN 1989B at the same phase. Middle below top: Comparison of the spectrum at 9 days past maximum of SN 1999ek with that of SN 1989B at about the same time. Middle above bottom: Corner plot with the posterior probability at 1 σ , 2 σ , 3 σ of distance and relative intrinsic reddening of SN 1999ek in relation to SN 1989B. Bottom: Corner plot with the posterior probability at 1 σ , 2 σ , 3 σ of distance moduli and relative intrinsic reddening of SN 1999ek in relation to SN 1989B.

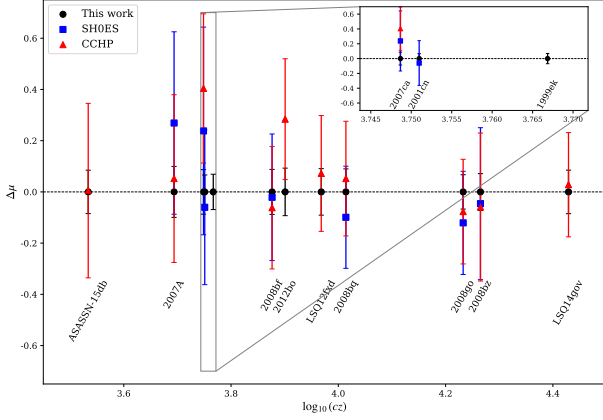


Figure 13. Difference in moduli of distance between the Pantheon+ SH0ES and those obtained in this work and between the CCHP and those obtained in this work. The points in black are those obtained in this work anchored in Cepheids, points in blue are from Pantheon+ SH0ES using Cepheids and triangles in red are Cepheids distances from CCHP. Some results in blue are not aligned: those are the ones by Pantheon+ for distances where the color c of those SNe Ia is underestimated, in particular for SN 2007A, SN 2007ca. The red triangles which are not aligned, corresponding to SNe Ia such as SN 2007ca, arise in an underestimate of the reddening from the CCHP. There is also the misalignment of SN 2012bo linked to its parameterization. A view of the results by Pantheon+ and by the CCHP compared with the “SNe Ia twins” results can be seen in Figures 14 and 15 for the sake of a better understanding.

We have modeled the statistical errors in the distance moduli for our method adding the various contributions. We have added in quadrature (see 4) the error in the fit of the distance modulus obtained with the MCMC, σ_{fit} , and the covariance between the modulus and the $\Delta(E(B-V))$, $\text{cov}(\mu, E(B-V))$. We have as well the error due to a difference of stretch between the Hubble flow SNe Ia and their nearby twins that goes into the distance modulus $\sigma^2(\mu, \Delta s_{BV})$. We model it taking into account the CSP impact of the error in stretch in the distance moduli of the SNe Ia. We must say that, in our method, twins at low z and in the Hubble flow have very close stretches: within ~ 0.02 in s_{BV} . The light curves agree so well between low z SNe Ia and their twins in the Hubble flow (once one makes the cosmological $(1+z)$ time broadening and magnitude dilution) that, in their plot of one over the other, they are often indistinguishable (see Appendix A). We add as well in quadrature the estimated intrinsic error σ_{int} (see Appendix B).

The error budget in μ is:

$$\sigma_{\mu}^2 = \sigma_{\text{fit}}^2 + 2\text{cov}(\mu, E(B-V)) + \sigma^2(\mu, \Delta s_{BV}) + \sigma_{\text{int}}^2 \quad (4)$$

15

where Δs_{BV} is the difference of stretch between the SNe Ia in the Hubble flow and in the low z twins. σ_{int} is the intrinsic error due to intrinsic uncertainties in flux comparison (see Methodology section).

For the H_0 error calculation, we incorporate the error in peculiar velocity affecting the cosmological z value. This error in the peculiar velocity is not attributed to the distance modulus because it enters into the H_0 calculation rather than in the distance. We adopt an error of 300 km s^{-1} in the calculation of the cosmological redshift (the cosmological redshifts are adopted from NASA/IPAC Extragalactic Database). According to authors who model peculiar velocities for the fields of Pantheon+ (Carr et al. 2022), those peculiar velocities in galaxies can reach to 250 km s^{-1} , and 100 km s^{-1} for group of galaxies (Carrick et al. 2015). For this reason, 300 km s^{-1} is a conservative estimate of the error.

This error on H_0 and those from the terms that enter in (5, see below) are added to the error budget for H_0 .

We obtain the individual values of H_0 for each measured SN Ia.

To calculate the value of H_0 we use the expression valid for a flat cosmology, $\Omega_k = 0$:

$$H_0 = \frac{cz}{D} \left(\frac{1 + z_{\text{hel}}}{1 + z} \right) \left[1 + \frac{1}{2}(1 - q_0)z - \frac{1}{6}(1 - q_0 - 3q_0^2 + j_0)z^2 + \mathcal{O}(z^3) \right] \quad (5)$$

where H_0 is the Hubble constant, D is the distance to the SN Ia, c is the speed of light, z is the cosmological redshift, z_{hel} the heliocentric redshift. The first and second order terms multiplied by z in the parenthesis involve the deceleration parameter, q_0 , and the cosmic jerk, j_0 . $q_0 = \Omega_m/2 - \Omega_{de}$ which taking $\Omega_m = 0.3$ give $q_0 = -0.55$ ($\Omega_{de} = 1 - \Omega_m$ for a flat cosmology). The cosmic jerk can be taken to be $j_0 = 1$ for the range of redshifts that we are using.

Our agreement (Table 5) with the distances using Cepheids in B by the CCHP is consistent with the findings of Uddin et al. (2024). Those authors obtained $H_0 = 72.37 \pm 0.71 \text{ km s}^{-1} \text{ Mpc}^{-1}$. Concerning TRGB in

¹⁵ $\sigma^2(\mu, \Delta s_{BV}) = (P1 + 2P2(\Delta s_{BV} - 1))^2 \sigma(\Delta s_{BV})^2$, see (7) for $P1, P2$.

Table 5. Comparison of distance moduli and H_0 with Pantheon+ and CCHP results^a

SN	z_{hel}	This work			SH0ES		CCHP
		μ	H_0	μ	H_0	μ	H_0
2012 bo	0.0265	35.162±0.0453	75.291±3.243	...	...	35.446±0.217	66.040±6.599
2008bq	0.0340	35.827±0.0739	72.419±3.240	35.728±0.178	75.796±5.964	35.879±0.205	70.705±8.132
2008bz	0.0603	37.214±0.0714	69.422± 2.547	37.168±0.288	71.072±3.945	37.155±0.280	71.334±9.198
LSQ12fxd	0.0312	35.543±0.0828	74.108± 3.702	—	—	- 35.615±0.207	72.497±6.911
2008bf	0.0235	35.073±0.112	74.085±4.820	35.052±0.231	74.716±7.494	35.011±0.222	76.192±7.789
2007A	0.0176	34.265±0.140	70.267± 6.223	34.534±0.342	62.043±9.041	34.317±0.312	68.604±9.857
ASASSN-15db	0.0109	33.527±0.0834	67.913±6.507	—	—	33.532±0.330	67.757±10.297
LSQ14gov	0.0896	37.912±0.0539	73.594 ±2.009	—	—	37.940±0.185	73.732±6.336
SN 2007ca	0.0140	33.940±0.0524	74.440±5.256	34.178±0.396	66.709±13.378	34.344±0.278	61.802±7.912
SN	This work z	Cepheids			CCHP		H_0
		μ	H_0	μ	H_0	μ	
2001cn	0.0156	33.947±0.0584	74.362±5.321	33.887±0.2949	76.421 ± 11.119	—	—
2008go	0.0623	37.081±0.277	72.284±9.299	36.96±0.19	77.955±7.128	37.004±0.193	76.392±6.790
1999ek	0.0177	34.342±0.0664	72.611±4.677	—	—	—	—

^aThis calibration uses the Cepheids distances to the low- z anchor galaxies.**Table 6.** Galactic and total reddenings of the SNe Ia in the Hubble flow

Supernova	$E(B - V)_{MW}$	$\Delta E(B - V)_{Host}$	$E(B - V)_{Tot}$	c	$B - V$
SN 2012bo	0.046	0.000 ± 0.004	0.046±0.004	-	0.012±0.006
SN 2008bq	0.060	0.000 ± 0.003	0.060±0.003	0.013±0.025 ⁵	0.045±0.009
SN 2008bz	0.023	0.00± 0.022	0.023 ^{+0.022} _{-0.000}	-0.136±0.023 ⁵	-0.006±0.006
...	...	...	...	-0.137±0.036 ⁶⁵	...
LSQ12fxd	0.022	0.000±0.002	0.022±0.002	-	-0.006±0.006
SN 2008bf	0.036	0.015± 0.007	0.051±0.007	-0.149±0.022 ⁵	-0.046±0.006
...	...	...	...	-0.149±0.026 ⁵⁷	...
...	...	...	...	-0.056±0.026 ⁶⁴	...
SN 2007A	0.074	0.209± 0.012	0.283±0.012	0.117±0.025 ⁵	0.201±0.010
...	...	...	...	0.063±0.033 ⁶⁵	...
ASASSN-15db	0.03	0.026±0.008	0.056±0.008	-	0.029±0.003
LSQ14gov	0.0377	0.015±0.001	0.0527±0.001	-	-0.104±0.008
SN 2007ca	0.067	0.199 ± 0.001	0.266±0.001	0.204±0.024 ⁵	0.260±0.010
...	...	...	...	0.151±0.023 ⁵⁷	...
...	...	...	...	0.207±0.028 ⁶⁴	...
SN 2001cn	0.052	0.133 ± 0.03	0.183±0.03	0.118±0.027	-
SN 2008go	0.033	0.020 ± 0.01	0.053±0.01	-0.016±0.023 ⁵	0.042±0.010
...	...	...	...	-0.007±0.035 ⁵¹	...
SN 1999ek	0.502	0.22 ± 0.006	0.722±0.006	-	-
⁵ CSP	⁵⁰ LOWZ/JRK07	⁵¹ LOSS1	⁵⁷ LOSS2	⁶⁴ CfA3K	⁶⁵ CfA4p2

B, Uddin et al. (2024) gave $H_0 = 70.25 \pm 0.71 \text{ km s}^{-1} \text{ Mpc}^{-1}$. And for *SBF*, (Surface Brightness Fluctuations) Uddin et al. (2024) obtained $H_0 = 72.45 \pm 0.94$.

6.2. Limitations of the parameterizations

Pantheon+ obtains the SNe Ia distances in the Hubble flow through a process that implies describing the SNe Ia with SALT2. The CCHP also makes this three rung ladder parameterizing the light curve-rate of decline relation but in a different way.

In this section, we analyse some shortcomings that we have found in the parameterization to obtain the distance moduli through the light curves using either SALT2 or the CSP new light curve-rate of decline relation.

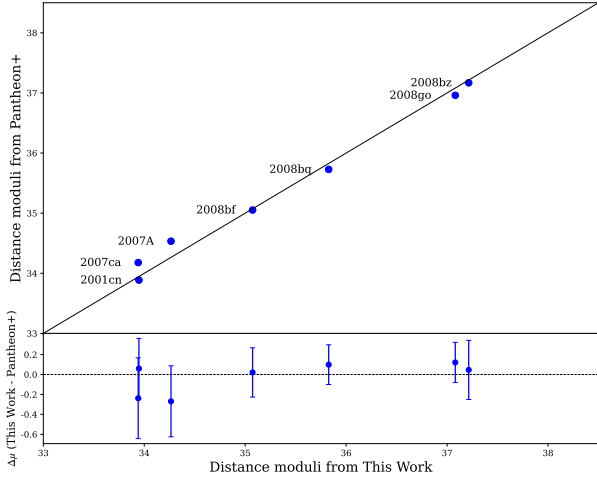


Figure 14. Top panel: Distance moduli in this work a versus those of Pantheon+. The distances span a range from 60 Mpc to 400 Mpc. Here we can see how for the Pantheon+ SN 2007A, SN 2007ca stand out due to a too low color 'c' estimation.

The Pantheon+ light curves are parameterized through the SALT2 light-curve fitting (Tripp 1998; Brout et al. 2022). In this approach the distance modulus is defined as:

$$\mu = m_B + \alpha x_1 + \beta c + M + \delta_{B,a} + \delta_{Host} \quad (6)$$

where α and β are global nuisance parameters relating stretch x_1 and the color c (respectively) to luminosity. M is the fiducial absolute magnitude of an SN Ia, which can be calibrated by setting an absolute distance scale with

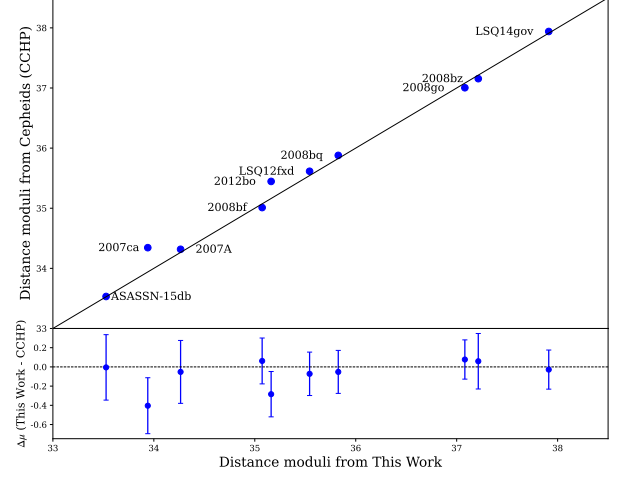


Figure 15. Top panel: Distance moduli in this work a versus those by the CCHP (Cepheids). The distances span a range from 60 Mpc to 400 Mpc. Here we can see how SN 2007ca stands out due to a too low correction by BV in relation to E(B-V) estimation. There is an error in the SN 2012bo parameters for the light curve-rate of decline correction.

primary distance anchors such as Cepheids. The $\delta_{B,a}$ bias is a correction term to account for selection biases that is determined from simulations following Popovic et al. (2021), δ_{Host} is the luminosity correction (step) for residual correlations between the standardized brightness of a SN Ia and the host-galaxy mass.

The relation used by the CSP and CCHP instead of being linear in stretch (like in Tripp 1998), is quadratic. The reddening correction is simply a constant, β , multiplied by the observed color of the SNe Ia, $(B-V)$ where B and V are the apparent, K-corrected peak magnitudes. P1 is the linear coefficient and P2 is the quadratic coefficient in $(s_{BV}-1)$, which encapsulates the Phillips relation in the s_{BV} stretch computed by SNOoPy; α_M is the slope of the correlation between peak luminosity and host stellar mass M_* (see for instance Freedman et al. 2019).

$$\mu_{\text{obs}} = m_x - P0 - P1(s_{BV} - 1) - P2(s_{BV} - 1)^2 - \beta(B - V) - \alpha_M(\log_{10} M_*/M_\odot - M_0) \quad (7)$$

We have noticed that, for reddened SNe Ia, the two parameterizations of the stretch fail to obtain good 'color' values and this impacts in the distance to SNe Ia. For instance, within the SALT2 parametrization, the color 'c' in (5) obtained by Pantheon+ tends to be low in the reddened SNe Ia in the Hubble flow (see Figure 14).

Table 7. Distance factors relative to anchors

SN	Distance factor	Anchor
2012bo	5.180 ± 0.017	2013dy
2008bq	7.038 ± 0.033	2013dy
2008bz	40.291 ± 0.001	2011fe
LSQ12fxd	6.175 ± 0.019	2013dy
2008bf	8.030 ± 0.098	2013aa
2007A	5.532 ± 0.102	2013aa
ASASSN-15db	7.376 ± 0.105	2011fe
LSQ14gov	29.674 ± 0.031	2013aa
SN2007ca	2.966 ± 0.007	2013dy
SN 2001cn	5.563 ± 0.040	1989B
SN 2008go	23.564 ± 0.195	1989B
SN 1999ek	6.675 ± 0.011	1989B

Such too low 'c' then reflects in a larger value for the distance modulus and carries into the Hubble flow values of the order of 60 to 65 km s⁻¹ Mpc⁻¹ (Table 5).

This happens as well with the parametrization by the CCHP (see 6), despite the fact that there is a term in (B-V). (B-V) acts as a color term that would include the presence of reddening. In fact, the BV color in the abovementioned CSP parameterization in (5) would be the combination of the intrinsic color of the SN Ia at maximum and the E(B-V) reddening. At maximum the intrinsic ($B_{max} - V_{max}$) of SNe Ia can be slightly negative, if $\Delta m(B)_{15}$ is lower than 1.1 (see expression 7 in Phillips et al. 1999)¹⁶. In Table 6, we include the $E(B-V)_{MW}$ from Schlafly & Finkbeiner (2011), the derived $E(B-V)_{Tot}$ from the spectra, the 'c' parameter for this SN Ia in the Pantheon+ compilation, and the 'BV' parameter in the CCHP compilation.

As mentioned in Freedman et al. (2019), there is an alternative parameterization by Burns et al. (2014) where instead of the $\beta(B-V)$ term there is a $R_B E(B-V)$ term to solve better the extinction. However, that has not been implemented in the present derivation of distance moduli for the CSP sample. So, this is the limitation we are seeing.

SN 2007A has a wrong value of its color 'c'. That causes a too large distance and individual point for H_0 with low values. It does not have a correct distance modulus in Pantheon+, as it has a color parameter in two different entries of $c = 0.117 \pm 0.024$ and $c = 0.067 \pm 0.033$. We have found and $E(B-V)_{Tot} = 0.283 \pm 0.012$. The CSP gives a value of $BV = 0.201 \pm 0.01$. The CSP gets a consistent distance modulus in this case as its 'BV' value would coincide with an $E(B-V)$ from 0.283 ± 0.012 plus a small negative value for the intrinsic color. The Pantheon+ does not provide a consistent distant modulus.

SN 2007ca requires a larger color correction than presented in Pantheon+. The color correction in Pantheon+, where it has 3 different entries for the SALT2 parameters is $c = 0.151 \pm 0.023$, $c = 0.204 \pm 0.023$ and $c = 0.207 \pm 0.028$. The $E(B-V)$ for this SN Ia is already 0.266 ± 0.001 . The CSP gives as its BV parameter 0.26 ± 0.01 , and this time it falls a little short.

On the other hand, other SNe Ia have a consistent 'c' or 'BV' values. For instance, in SN 2001cn, Pantheon+

gives a color with 'c' = 0.118 ± 0.027 . The spectra show that $E(B-V)$ is 0.183 ± 0.005 . Given that c should reflect the intrinsic color of SNe Ia, which is $(B_{max,int} - V_{max,int}) = -0.072 (\pm 0.049)$ mag for this SN Ia, according to the expression (7) in Phillips et al (1999), if we add the reddening the final color 'c' is 0.111 ± 0.079 . Here the final distances are similar, but the one by Pantheon+ has very large errors, that might be linked to not have used the light curves for this SN Ia (Krisciunas et al. 2004) that would allow a very good x1 and c determination.

Table 8. H_0 and distances calibrated with JAGB^a

SN	μ	H_0
2012bo	35.162 ± 0.0453	75.291 ± 3.243
2008bq	35.827 ± 0.0739	72.419 ± 3.240
2008bz	37.236 ± 0.0714	68.722 ± 2.522
LSQ12fxd	35.543 ± 0.0828	74.108 ± 3.702
2008bf	35.103 ± 0.112	73.069 ± 4.767
2007A	34.294 ± 0.140	69.335 ± 6.095
ASASSN-15db	33.549 ± 0.0834	67.228 ± 6.477
LSQ14gov	37.942 ± 0.0539	72.585 ± 2.009
2007ca	33.940 ± 0.0524	74.440 ± 5.256
SN	μ	H_0
2001cn	33.947 ± 0.0584	74.362 ± 5.321
2008go	37.081 ± 0.277	72.284 ± 9.299
1999ek	34.342 ± 0.0664	72.611 ± 4.677

^aThe distances are anchored using JAGB by the CCHP for the 9 SNe Ia in the upper part of the Table and using TRGB by the CCHP in the last three SNe Ia of the Table.

6.3. The value of H_0

Our results for individual SNe Ia, in Tables 5 and 8 clearly confirm that the Hubble tension is real, since they are incompatible with the H_0 value derived from the CMB. The results of the Hubble diagram, when the SNe Ia are anchored in JAGBs (CCHP) (Table 8), are quite similar to the anchoring in Cepheids.

¹⁶ $B_{max} - V_{max} = -0.070 (\pm 0.012) + 0.114 (\pm 0.037) \times (\Delta m_{15}(B) - 1.1)$, $\sigma = 0.0042$

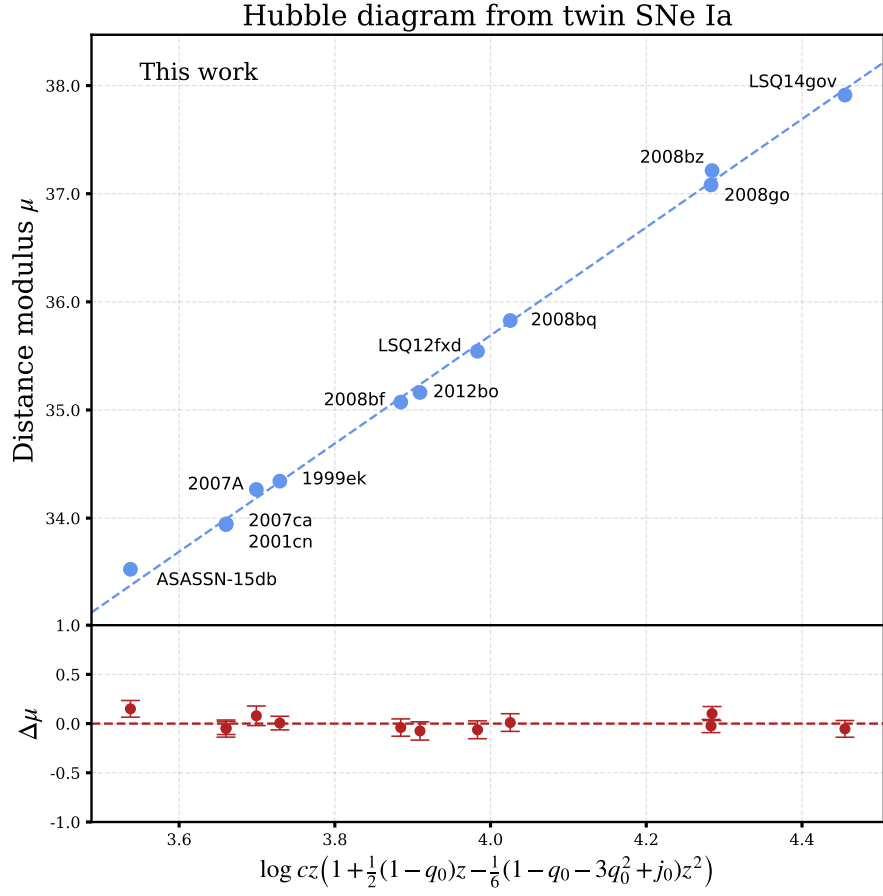


Figure 16. Hubble diagram with the distances obtained in this work. The intercept of the straight line with the y-axis corresponds to $H_0 = 72.79 \pm 0.78 \text{ km s}^{-1} \text{ Mpc}^{-1}$.

Table 9. Comparison of JAGB distances (CCHP) with JAGB distances (SH0ES²)

Galaxy	μ_{JAGB} (CCHP) ¹ (mag)	μ_{JAGB} (SH0ES) ² (mag)
M101	29.21±0.03 (stat)±0.03 (sys)	29.12±0.06
NGC 1365	31.38±0.02 (stat)±0.03 (sys)	31.34±0.04
NGC 2442	31.60±0.01 (stat)±0.04 (sys)	31.56±0.05
NGC 4536	30.97±0.01 (stat)±0.03 (sys)	30.93±0.04
NGC 4639	31.73±0.02 (stat)±0.03 (sys)	31.69±0.04
NGC 5643	30.58±0.01 (stat)±0.04 (sys)	30.54±0.04
NGC 7250	31.59±0.02 (stat)±0.04 (sys)	31.59±0.04
NGC 3972	31.67±0.04 (stat)±0.03 (sys)	—
NGC 4038	31.53±0.06 (stat)±0.04 (sys)	—
NGC 4424	31.15±0.04 (stat)±0.03 (sys)	—

¹Lee et al. (2025). ²Li et al. (2025)

The value, for the 12 SNe Ia, is $H_0 = 72.56 \pm 1.54$ (stat) ± 1.33 (syst) $\text{km s}^{-1} \text{Mpc}^{-1}$ when calibrating the anchor distances with Cepheids (by SH0ES) and 72.20 ± 1.53 (stat) ± 1.33 (sys) $\text{km s}^{-1} \text{Mpc}^{-1}$ when calibrating with JAGB (by the CCHP). The average is $H_0 = 72.38 \pm 1.54$ (stat) ± 1.33 (sys) $\text{km s}^{-1} \text{Mpc}^{-1}$, far from the $67.4 \pm 0.5 \text{ km s}^{-1} \text{Mpc}^{-1}$ of the Planck Collaboration.

Figure 16 is the Hubble diagram of the SNe Ia in this work anchored in Cepheids (Table 5). The intercept corresponds to $H_0 = 72.79 \pm 0.78 \text{ km s}^{-1} \text{Mpc}^{-1}$.

If we take the SNe Ia of the BL class only, for which the anchor distance is known with the greatest precision (coincidence between CCHP distances and Saha et al. 1999), the simple mean is $H_0 = 73.00 \pm 3.98$ (stat) $\text{km s}^{-1} \text{Mpc}^{-1}$ and the weighted average from the three SN Ia 73.24 ± 3.29 (stat) $\text{km s}^{-1} \text{Mpc}^{-1}$.

Our results from comparing the spectra of very nearby SNe Ia with those of their twins in the Hubble flow also validates the three rung procedures used by the Pantheon+ and CCHP collaborations, even at the level of individual SNe Ia, since the discrepancies can easily be explained by differences in the evaluation of the total reddenings or the different light-curve parameters adopted, as we have seen in the previous Section.

To leave our results stable against future changes in the distances of the anchors in the nearby galaxies, we publish the distance factors between the anchors and the SNe Ia distances. The factors should be multiplied by the distances of the anchors to obtain the distances to the SNe Ia in the Hubble flow (Table 7).

Concerning the different calibrations used for setting our anchor distances, we see that the distance moduli from TRGB by the CCHP have fluctuated during the

last year. In addition, their distances are very different when using the same data but analysed by other groups (Anand et al. 2022).

Consequently, we have obtained the two H_0 values given above, for the two different calibrations of our anchors using Cepheids and JAGBs (Table 5 and Table 8).

It is worth to remark the distance to M101 measured from SH0ES with Cepheids and with the JAGB stars from both the CCHP and SH0ES. Those distance values clearly favor a value below $70 \text{ km s}^{-1} \text{Mpc}^{-1}$ for H_0 from SNe Ia twins of SN 2011fe. Table 9 illustrates the difference in distance moduli to M101 obtained by the two collaborations with the same method: it is a difference of $\Delta\mu_{\text{M101}} = 0.09 \text{ mag}$, which translates in $\sim 4 \text{ km s}^{-1} \text{Mpc}^{-1}$ for ΔH_0 . If our sample would be dominated by 2011fe-like SNe Ia, the total H_0 would be lower than what it is. However, the other values of distances using JAGB stars for the rest of galaxies differ only by 0.04 mag between the larger distance value by CCHP and the shorter one from SH0ES. As the sample contains SNe Ia twins from SN 2013aa and SN 2013dy, the values above $70 \text{ km s}^{-1} \text{Mpc}^{-1}$ dominate. We pointed, in RLGH24, a preference for the distance to SN 2011fe from the spectra of $\sim 6.5 \text{ Mpc}$, in agreement with the value obtained from Blue Supergiants (Bresolin et al. 2022) of $6.5 \pm 0.2 \text{ Mpc}$. Inevitable, using the 10 galaxy calibrators by the JAGB in rung 2, as it is done by the CCHP, could bring H_0 to lower values than $70 \text{ km s}^{-1} \text{Mpc}^{-1}$ given the calibration of M101. That would depend on the weight of M101 in the whole sample. However, it is surprising such low mean value of $H_0 \sim 67 \text{ km s}^{-1} \text{Mpc}^{-1}$ published by Freedman et al. (2025). The case deserves further exploration.

7. SUMMARY AND CONCLUSIONS

The use of the "SNe Ia twins" to obtain H_0 has revealed that the Hubble tension is real. The distances obtained through this direct path go from $\sim 6 \text{ Mpc}$ to around 300 Mpc in a single step.

Comparing our distances with those in the Pantheon+ and CCHP samples using Cepheids, we find a general agreement, with a few exceptions, mostly related with the determination of the reddening of some SNe Ia. The results in the Hubble diagram coincide and the values of H_0 are very close.

This implies that the SNe Ia calibrators in rung 2 are a good sample that achieves a consistent final value of

H_0 in rung 3. The three-rung method seems to work. It could fail if a biased sample in rung 2 were taken. Then, the final value of H_0 would be distorted, and the results of the Hubble diagram changed. The large samples in Pantheon+ or CCHP using Cepheids as calibrators have values of individual H_0 whose average give close values for H_0 . Using the TRGB obtained by the CCHP, the distances of the galaxies in rung 2 would be larger and the H_0 smaller, in contrast to the TRGB from SH0ES. A measurement that uses Cepheids from SH0ES and JAGB from the CCHP with the twins method crosses the aisle between the two collaborations. The opportunity comes from similar distance values for a few galaxies below 20 Mpc.

At the individual level, the distances to each SN Ia are affected by the limitations of the parameterization of the light curves used by the two collaborations. The method of twin SNe Ia, with its use of spectra, is very powerful in getting the reddening of the SNe Ia in the Hubble flow. An approach as that developed here can point to the failures in the distance moduli for these supernovae that give H_0 in the extremes of the distribution (very low or very large H_0). SNe Ia with color larger than 'c' = 0.1 in the Pantheon+ sample give rise to low reddening correction and larger distance moduli. This leads to individual points of H_0 falling out of the range of discussion, i.e., in the tail of H_0 below $65 \text{ km s}^{-1} \text{ Mpc}^{-1}$. A similar effect is seen with the BV color. Those tails come from SNe Ia where the color terms 'c', or '(BV)' are not well estimated or even the rates of decline of the lightcurves parameterized in the form of 'x1' in SALT2 or s_{BV} in CCHP have been wrongly estimated. Sometimes large uncertainties are quoted to accompany those values, but in some cases they are not. Those extreme values do not alter significantly the mean for H_0 . However, for those particular SNe Ia in the Hubble flow the distance remains unknown.

Inaccurate point values in distances and therefore in H_0 individual points (even when enclosing these points within a large error bar) would interfere with the proper determination of possible cosmic bulk flow in the z range $0.015 < z < 0.1$. An important task is to improve the Pantheon+ and CSP distances. In this sense, our approach can help a lot to give the right distances and right individual H_0 values.

The use of SN 1989B in M66 as anchor has been very fruitful. It has allowed to get accurate distances to three SNe Ia: SN 2001cn, SN 2008go and SN 1999ek. As said before, the distance to M66 with TRGB by Freedman

et al. (2019) (obtained by Hoyt et al. 2019) coincides with the Cepheids value from Saha et al. (1999).

We finally stress that the present use of two different calibrations, that with Cepheids from Pantheon+ and the one with JAGB from CCHP, do converge to the same result. This concordance found using anchors from the two collaborations rarely happens and looks very promising. There is a very good agreement on the distances to NGC 7250 and NGC 5643 derived with Cepheids by SH0ES and derived with the use of J-Asymptotic Giant Branch stars (JAGB stars) by the CCHP. The difference for SNe Ia in the Hubble flow anchored in one or the other way is negligible: our SNe Ia sample in the Hubble flow anchored in Cepheids gives $H_0 = 72.56 \pm 1.54 \text{ (stat)} \pm 1.33 \text{ (sys)}$ and anchored in JAGB stars gives $H_0 = 72.20 \pm 1.53 \text{ (stat)} \pm 1.33 \text{ (sys)}$. The mean would be $H_0 = 72.38 \pm 1.54 \text{ (stat)} \pm 1.33 \text{ (sys)}$.

This work has made use of spectra from the *Weizmann Interactive Supernova data REpository* (WIS-eREP), spectra and photometry of the Carnegie Supernova Project (CSP). The authors acknowledge Pantheon+ and the CSP for opening their data releases and software to the community. We are very grateful to Nidia Morrell (from the CSP) for exchanges in relation to the photometry of CSP SNe Ia. We thank the referee for his careful and very helpful report. PR-L and A.Q-E acknowledge support from grant PID2021-123528NB-I00, from the Spanish Ministry of Science and Innovation (MICINN). JIGH acknowledges financial support from MICIU grant PID2023-149982NB-I00. A.P. acknowledges support from the PRIN-INAF 2022 project "Shedding light on the nature of gap transients: from the observations to the model".

Software: AstroPy (Price-Whelan et al. 2018); H0CSP.ipynb (Uddin et al. 2024); Matplotlib (version 3.2.1, Hunter 2007); NumPy (version 1.18.2, Oliphant 2006; van der Walt et al. 2011). SNooPy (Burns et al. 2011);

REFERENCES

- Anand, G.S., Tully, R.B., Rizzi, L., Riess, A.G., & Yang, W. 2022, ApJ, 932, 15
- Boone, K., Fakhouri, H., Aldering, G.A., et al. 2016, AAS Meeting, 227, 237.101
- Branch, D., Dang, L.C., Hall, N., et al. 2006, PASP, 118, 560
- Bresolin, F., Kudritzki, R.P., Urbaneja, M.A., Sextl, E., & Riess, A.G. 2025, ApJ, 991, 151

- Brout, D., Taylor, G., Scolnic, D., et al. 2022, *ApJ*, 938, 111
- Burns, C.R., Stritzinger, M., & Phillips, M.M. 2011, *AJ*, 141, 19
- Burns, C.R., Stritzinger, M., Phillips, M.M., et al. 2014, *ApJ*, 789, 32
- Burns, C.R., Ashall, C., Contreras, C., et al. 2020, *ApJ*, 895, 11J8
- Burrow, A., Baron, E., Ashall, et al. 2020, *ApJ*, 901, 154
- Cardelli, J. A., Clayton, G. C., & Mathis, J. S. 1989, *ApJ*, 345, 245
- Carr, A., Davis, T.M., Scolnic, D, et al. 2022, *PASA*, 39, e046
- Carrick, J., Turnbull, S.J., Lavaux, G., & Hudson, M.J. 2015, *MNRAS*, 450, 317
- Casper, C., Zheng, W., Li, W., et al. 2013, *CBET*, 3588
- Cenko, S.B., Thomas, R.C., Nugent, P.E., et al. 2011, *ATel*, 3583, 1
- Chassagne, R., & Santallo, R. 2001, *IAUC*, 7643
- Di Valentino, E., Mena, O., Pan, S., et al. 2021a, *CQ-Gra*, 38, 153001
- Di Valentino, E., Melchiorri, A., & Silk, J. 2021b, *ApJL*, 908, L9
- Di Valentino, E., Levi Said, J., & Sandakis, E.N. 2025a, *arXiv:2509.25288*
- Di Valentino, E., Said, J.L., Riess, A.G., et al. 2025b, *PDU*, 49, 101965
- Ducroft, M., Leget, P.F., Chotard, N., et al. 2014, *ATel*, 6859
- Fakhouri, H.K., Boone, K., Aldering, G., et al. 2015, *ApJ*, 815, 58
- Folatelli, G., Morrell, N., Phillips, M.M., et al. 2013, *ApJ*, 773, 53
- Foreman-Mackey, D., Hoog, D.W., Lang, D., & Goodman, J. 2013, *PASP*, 125, 300
- Freedman, W.L., Madore, B.F., Hatt, D., et al. 2019, *ApJ*, 882, 34
- Freedman, W.L., Madore, B.F., Hoyt, T.J., Jang, I.S., Lee, A.J., & Owens, K.A. 2024, *arXiv:2408.06153*
- Freedman, W.L., Madore, B.F., Hoyt, T.J., Jang, I.S., Lee, A.J., & Owens, K.A. 2025, *ApJ*, 985, 203
- Galbany, L., de Jaeger, T., Riess, A.G., et al. 2023, *A&A*, 679, A95
- Garnavich, P., Wood, C.M., Milne, P., et al. 2023, *ApJ*, 953, 35
- Gelman, A., Roberts, G.O., & Gilks, W. 1996, *Bayesian Statistics*,
- Goodman, J., & Weare, J. 2010, *Communications in Applied Mathematics and Computational Science*, 5, 65
- Griffith, C., Li, W., Cenko, S.B., & Filippenko, A.V. 2008, *CBET*, 1553
- Holoien, T.W.S., Stanek, K.Z., Kochanek, C.S., et al. 2015, *ATel*, 7078
- Hoyt, T.J., Freedman, W.L., & Madore, G.F. 2019, *ApJ*, 882, 150
- Hoyt, T., J., Beaton, R.L., Freedman, W.L., et al. 2021, *ApJ*, 915, 34
- Hoyt, T.J., Jang, I.S., Freedman, W.L., Madore, B.F., Owens, K.A., & Lee, A.J. 2025, *arXiv:2503.11769*
- Huang, C.D., Yuan, W., Riess, A.G., et al. 2024, *ApJ*, 963, 83
- Hunter, J.D. 2007, *CSE*, 9, 3
- Itagaki, K., Nakano, S., Quimby, R., et al. 2007, *IAUC*, 8843
- Itagaki, K., Yusa, T., Noguchi, T., et al. 2012, *CBET*, 3079
- Jacobson-Galan, V.W., Dimitriadis, G., Foley, R.J., & Kirpatrick, C.D. 2018, *ApJ*, 857, 88
- Jensen, J.B., Blakeslee, J.P., Cantiello, M., et al. 2025, *ApJ*, 987, 87
- Johnson, R., & Li, W. 1999, *IAUC*, 7286
- Kamionkowski, M., & Riess, A.G. 2023, *ARNPS*, 73, 153
- Krisciunas, K., Suntzeff, N.B., Phillips, M.M., et al. 2004, *AJ*, 128, 3034
- Krisciunas, K., Contreras, C., Burns, C.R., et al. 2017, *AJ*, 154, 211
- Lee, A.J., Freedman, W.L., Madore, B.F., Jang, I.S., Owens, K.A. & Hoyt, T.J. 2025, *ApJ*, 985, 182
- Li, S., Riess, A.G., Casertano, S., et al. 2025, *ApJ*, 988, 97
- Luckas, P., Trondal, O., & Schwartz, M. 2008, *CBET*, 1328
- Morrell, N., Phillips, M.M., Folatelli, G., et al. 2024, *ApJ*, 967, 20
- Mörtl, E., & Dhawan, S. 2018, *JCAP*, 1809, 025
- Murakami, Y.S., Riess, A.G., Stahl, B.E., et al. 2023, *JCAP*, 11, 046
- Nugent, P.E., Sullivan, M., Cenko, S.B., et al. 2011, *Natur*, 480, 344
- Oliphant, T.E. 2007, *CSE*, 9, 10
- Pan, Y.C., Foley, R.J., Kromer, M., et al. 2015, *MNRAS*, 452, 4307
- Pansky, X., Li, W., & Filippenko, A.V. 2008, *CBET*, 1307
- Pantos, I., & Perivolaropoulos, L. 2026, *arXiv:2601.00650v2*
- Parrent, J.T., Sand, D., Valento, M., Graham, D.A., & Howell, D.A. 2013, *ATel*, 4817, 1
- Perlmutter, S., Aldering, G., Goldhaber, G., et al. 1999, *ApJ*, 517, 565
- Phillips, M.M. 1993, *ApJL*, 413, L105

Phillips, M. M., Lira, P., Suntzeff, N.B., Schommer, R. A., Hamuy, M., & Maza, J. 1999, *AJ*, 118, 1766

Planck Collaboration 2020, *A&A*, 641, A6

Popovic, B., Brout, D., Kessler, R., Scolnic, D., & Lu, L. 2021, *ApJ*, 913, 49

Poulin, V., Smith, T.L., Calderón, R., & Simon, T. 2025, *PhRvD*, 111, 083552

Price-Whelan, A.M., Sipőcz, B.M., Günther, H.M., et al. 2018, *AJ*, 156, 123

Puckett, T., Orff, T., Newton, J., Madison, D., & Li, W. 2007, *CBET*, 795

Richmond, M.W., & Smith, H.A. 2012, *JAVSO*, 40, 872

Riess, A.G., Filippenko, A.V., Challis, P., et al. 1998, *AJ*, 116, 1009

Riess, A.G., Yuan, W., Macri, L.M., et al. 2022, *ApJ*, 934, L7

Riess, A.G., Li, S., Anand, G. et al. 2025, *ApJL*, 922, L34

Riess, A.G., Scolnic, D., Gagandeep, S., et al. 2025, *ApJ*, 977, 120

Rubin, D., Aldering, G., Betoule, M., et al. (2023, arXiv:2311.12098) 2025, *ApJ*, 986, 231

Ruiz-Lapuente, P., & González Hernández, J.I. 2024, *ApJ*, 977, 180

Ryan, S., & Visvanathan, R. 1989, *IAUC*, 4730

Saha, A., Sandage, A., Tammann, G.A., Labhardt, L., Machetto, F.D., & Panagia, N. 1999, *ApJ*, 522, 802

Schöneberg, N., Franco Abellán, G., Pérez Sánchez, A., Witte, S.J., Poulin, V., & Lesgourges, J. 2022, *PhR*, 984, 1

Scolnic, D., Brout, D., Carr, A., et al. 2022, *ApJ*, 938, 113

Schlaflly, E.F., & Finkbeiner, D.P. 2011, *ApJ*, 737, 103

Schneider, S.E., Thuan, T.X., Mangum, J.G., & Miller, J. 1992, *ApJS*, 81, 5

Stahl, B.E., Zheng, W., de Jaeger, T., et al. 2020, *MNRAS*, 492, 4325

Tartaglia, R., Sand, D., Wyatt, S., et al. 2017, *ATel*, 4158, 1

Tripp, R. 1998, *A&A*, 331, 815

Uddin, S.A., Burns, C.R., Phillips, M.M., et al. 2020, *ApJ*, 901, 143

Uddin, S.A., Burns, C.R., Phillips, M.M., et al. 2024, *ApJ*, 970, 72

Van Der Walt, S., Colbert, S.C., & Varoquaux, G. 2011, *CSE*, 13, 22

Vogl, C., Taubenberger, S., Csómier, G., et al. 2025, *A&A*, 702, A41

Waagen, E.O. 2013, *AAN*, 479, 1

Wells, L.A., Phillips, M.M., Suntzeff, B., et al. 1994, *AJ*, 108, 2233

Yang, Y., Hoeflich, P., Baade, D., et al. 2020, *ApJ*, 902, 46

Yuan, F., Quimby, R., Chamarro, D., et al. 2008, *CBET*, 1353

Zheng, W., Silverman, J.M., Filippenko, A.V., et al. 2013, *ApJL*, 788, L15

8. APPENDIX A: LIGHT CURVES OF SNE IA IN THE HUBBLE FLOW, COMPARED WITH THOSE OF THEIR LOW-REDSHIFT TWINS

In this Appendix, we show how the light curves of the SNe Ia in the Hubble flow from the CSP I coincide with those of our twins of reference, once these last ones are placed in the corresponding cosmological frame.

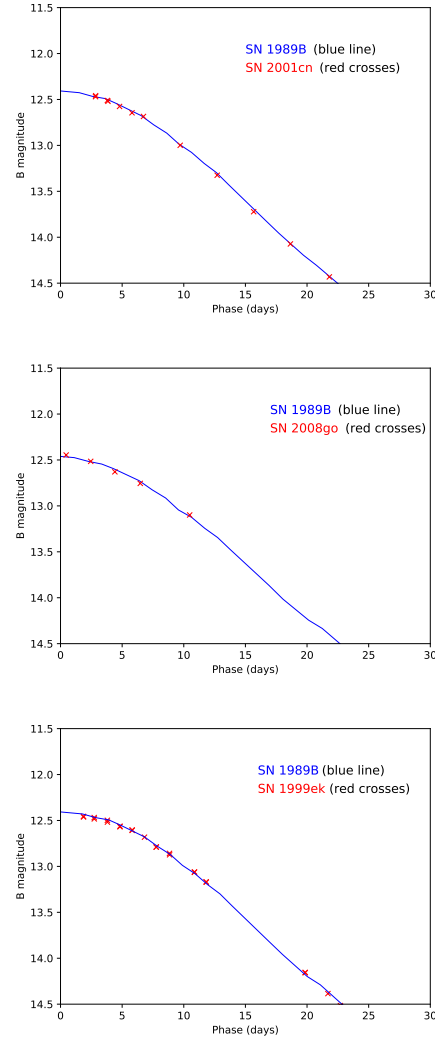


Figure 17. B light curves of the broad-line SNe Ia twins SN 2001cn/SN 1989B (left), SN 2008go/SN 1989B (middle) and SN 1999ek/SN 1989B (right).

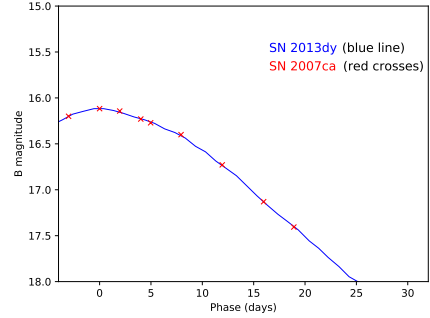
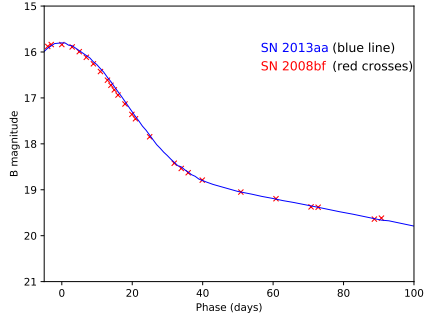
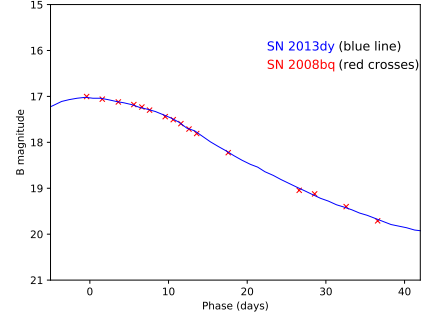
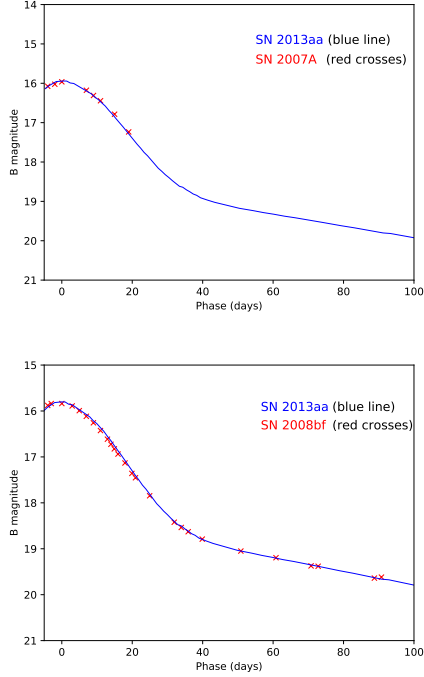


Figure 18. B light curves of the core-normal SNe Ia twins SN 2007A/SN 2013aa (left) and SN 2008bf/SN 2013aa (right).

9. APPENDIX B: INTRINSIC ERROR WITH THE “SNE IA TWINS” METHOD

In addition to the approach used in RLGH24 to estimate the error within the “twin SNe Ia” method, we have used in this work a refined estimation by introducing an additional term in the total error budget, namely the intrinsic scatter, σ_{int} , treated as a free parameter within the MCMC fit. This parameter is included in all fits and marginalized over. We present here the corner plots showing σ_{int} for two SNe Ia, SN 2001cn (Figure 20) and SN 1999ek (Figure 21).

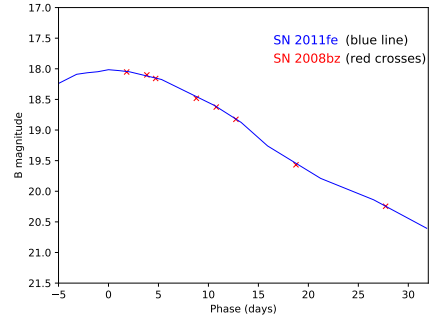


Figure 19. B light curves of the core-normal SNe Ia twins 2008bq/SN 2013dy (left), SN 2007ca/SN 2013dy (middle) and SN 2008bz/SN 2011fe (right) .

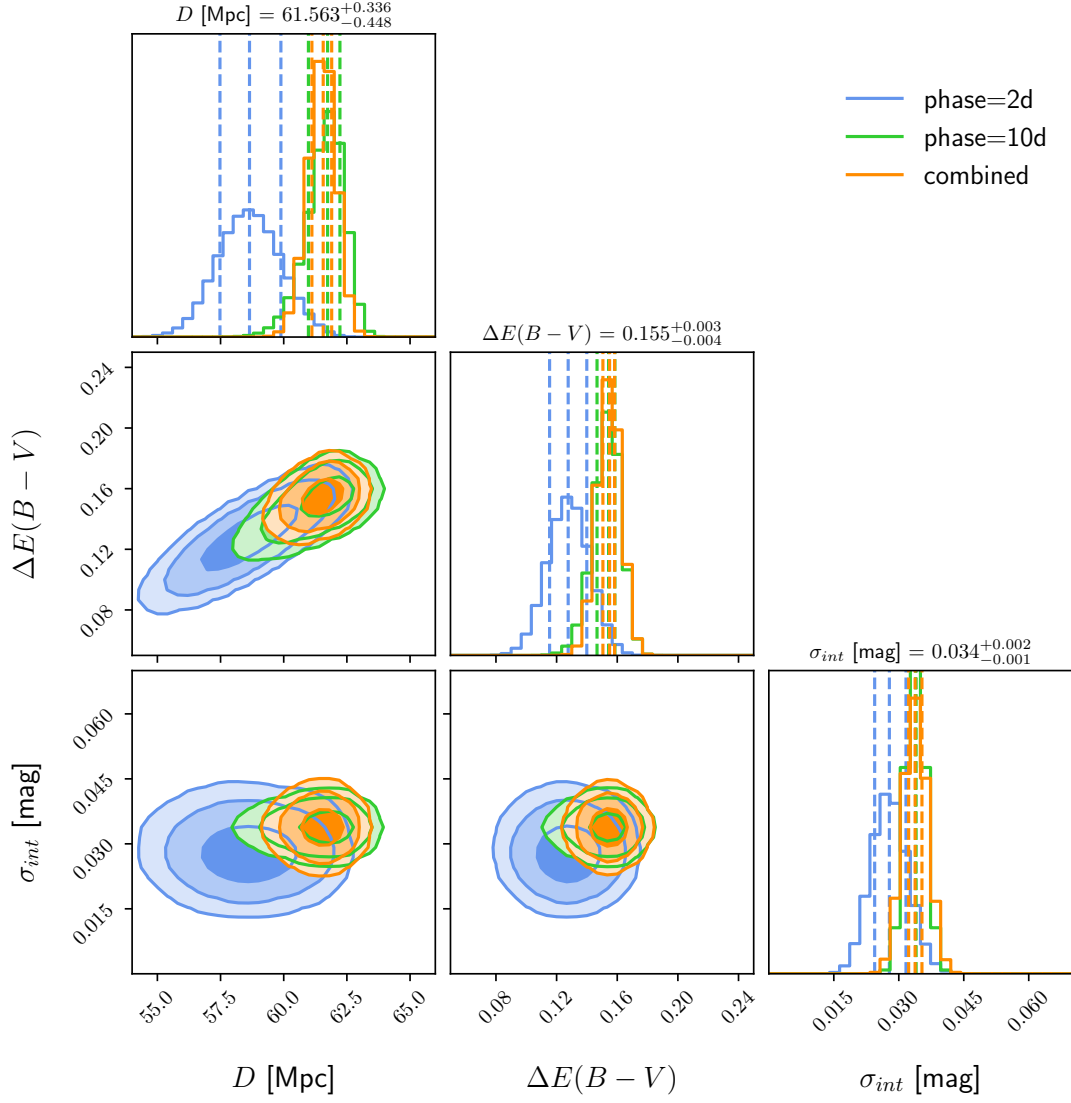


Figure 20. Corner plot with the posterior probability at 1σ , 2σ , 3σ of distance, relative intrinsic reddening of SN 2001cn in relation to SN 1989B and the intrinsic scatter magnitude σ_{int} of the method which results, in SN 2001cn, of $0.034^{+0.002}_{-0.001}$ mag.

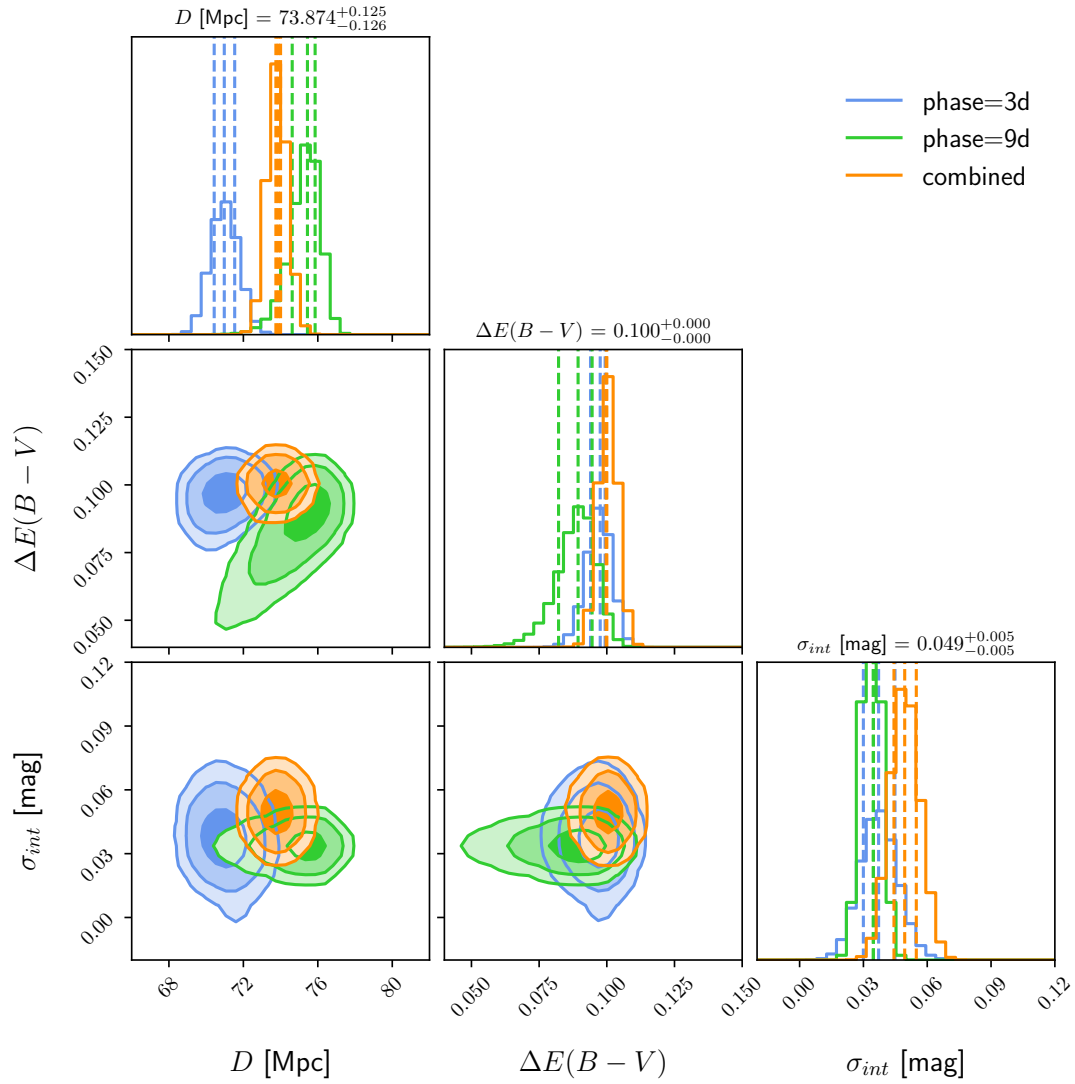


Figure 21. Corner plot with the posterior probability at 1σ , 2σ , 3σ of distance, relative intrinsic reddening of SN 1999ek in relation to SN 1989B and the intrinsic scatter magnitude σ_{int} of the method which results, in SN 1999ek, of 0.049 ± 0.005 mag.